



Developing two dynamic Malmquist-Luenberger productivity indices: An illustrated application for assessing productivity performance of Indian banks[☆]

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ABSTRACT

In the presence of undesirable output(s) in the production process, the Malmquist-Luenberger productivity index (*MLPI*) has emerged as a standard tool to measure total factor productivity change and its distinct components, namely efficiency change and technical change. Notwithstanding, the standard version of *MLPI* has certain genuine downsides, particularly the commonness of infeasibility while computing cross-period directional distance functions with negative data values and the black-box representation of the production technology. For overcoming these downsides, this paper advances two extensions of the standard *MLPI*. Specifically, we propose dynamic Malmquist-Luenberger and dynamic sequential Malmquist-Luenberger productivity indices to measure the productivity change in a system that can be represented by a dynamic network production structure. The proposed indices take a holistic view of production technology that connects several divisions internally by intermediate processes and use the carryovers flow over time to add a temporal dimension to it. To exhibit the capability of the proposed extensions of *MLPI*, we model a three-stage dynamic network structure for Indian banks and compute the indices of productivity change, efficiency change, and technical change for forty-two banks for the period from 2010 to 2017. The empirical findings elucidate that the observed productivity growth in the Indian banking industry is primarily driven by technical progress.

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1. Introduction

There is no denying of the fact that improvement in total factor productivity (TFP) is a *sine qua non* for the long-run sustainability of the firms. How well a firm stands in an environment of cut-throat competition hinges on how well the firm performs on the productivity front. Given the significance of productivity change, a large body of literature has developed, targeting at measuring the TFP growth rates using a range of productivity measurement approaches (see the excellent literature surveys of [18], and [2]). Among the recent methodological innovations that enticed numerous empirical applications are the Malmquist productivity index (*MPI*) and the Malmquist-Luenberger productivity index (*MLPI*). These indices allow decomposing the TFP change into its two

distinct drivers: efficiency change and technical change. Both of these indices are easy to calculate by applying either the parametric (like stochastic frontier analysis (SFA)) or non-parametric (like data envelopment analysis (DEA)) frontier approaches. The non-parametric DEA modeling lends an advantage by not imposing any prior functional form on the technology frontier. Unlike the parametric techniques that ask for an estimation of parameters and the probability distribution of the error measure (or noise term) in the cost/profit/production/distance frontier specification, the non-parametric DEA methodology relies upon the linear programming technique for obtaining empirical estimates of distance functions and calculating productivity indices based on those empirical estimates.

The *MPI*, as introduced by Caves et al. [23], computes the productivity change index using distance functions and is defined as a geometric mean of two adjacent periods' indices. Using directional distance functions (DDFs), Chung et al. [25] extended *MPI* to define *MLPI* for accommodating undesirable output(s) in computing productivity change. Oh and Heshmati [74] modified the conventional *MLPI* and developed the sequential Malmquist-

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Luenberger productivity index (*SMLPI*) based on the concept of the successive sequential production possibility set. Notwithstanding, the standard renditions of *MLPI* and *SMLPI* have two serious downsides: i) in the presence of negative data values, the researcher encounters an infeasibility problem while computing the cross-period DDFs under the constant returns-to-scale (CRS), and ii) it is entirely based on the black-box representation of the production technology. Thus, the standard *MLPI* does not recognize the dynamic network production structure and hence ignores the role of intermediate outputs and carryover activities. These noted downsides of *MLPI* and *SMLPI* serve as a motivation factor for this paper.

The main aim of this paper is to develop two novel extensions of *MLPI*, which we appropriately call as: i) dynamic Malmquist-Luenberger productivity index (*DMLPI*), and ii) dynamic sequential Malmquist-Luenberger productivity index (*DSMLPI*). The proposed productivity indices effectively overcome the above-mentioned flaws of standard versions of *MLPI* and *SMLPI*. The employability of the proposed productivity indices in the banking industry is provided for illustration purpose only since it is reasonable to expect negative values of some variables like net interest income and net profits for banking firms. It is worth mention here that the applicability of proposed indices could easily be generalized to any industry where one can open the black-box of production technology and segregate production activities into mutually exclusive distinct divisions or sub-processes that are linked via intermediate products. As opposed to the standard *MPI* and *MLPI* approaches, which overlook the internal sub-processes in the system and treat a decision-making unit (DMU) as the “black-box”, the proposed indices take a holistic view of the production structure by opening the black-box using a dynamic network production structure where multiple divisions in the DMU are connected vertically by means of intermediate activities, and multiple periods are connected through the carryovers. The proposed indices likewise circumvent the infeasibility problem, permit negative data, undesirable inputs and outputs, quasi-fixed inputs, and carryovers across time. The proposed *DMLPI* is based on contemporaneous benchmark technologies and allows the possibility of technical regress (i.e., inward movement of the frontier). Shestalova [86] points that i) technical regress appears to be conceivable for some industries such as the mining sector, and ii) technical regress and consequent productivity deterioration (if any) is usually a temporary (short-term) phenomenon. We, along these lines, argue that the use of *DMLPI* is suitable for measuring productivity growth for those samples having a small-time horizon. In addition, the proposed novel *DSMLPI*, other than carrying all the essential features of *DMLPI*, appropriately considers the progressive nature of technology (as the case in many industries, including banking) by enveloping all the preceding technologies in constructing sequential benchmark technologies as suggested by Tulkens and Vanden Eeckaut [96] and precludes any spurious technical regress. Following Oh et al. [75], we prescribe utilizing the proposed *DSMLPI* for measuring productivity change and its decomposed sources for the mid-term or long-term period rather than the short-term period.

One of the prime concerns that surfaced in the existing literature on the non-parametric measurement of TFP change is that in the presence of negative data, the infeasibility issue is very likely to emerge when computing DDFs under the CRS technology assumption. Portela and Thanassoulis [81] showed that the infeasibility issue gets resolved when *MLPI* is calculated under the variable returns-to-scale (VRS) assumption. Earlier, Grifell-Tatjé and Lovell [45], Lovell et al. [63], and Zofio [105] had raised an issue regarding the interpretation of productivity change under the VRS postulate. Notwithstanding in the extant literature, this issue is somewhat settled by Portela and Thanassoulis [81], who asserted that *MLPI* under VRS captures the productivity change net of

scale economies. In the present study, we measure the productivity change using both *DMLPI* and *DSMLPI* under the VRS technology assumption and decipher this change in the spirit of Portela and Thanassoulis [81].

Our paper adds value to the extant literature since we took a step forward on TFP measurement and augmented the standard *MLPI* index to define two new productivity indices, namely *DMLPI* and *DSMLPI*, to measure the productivity change in a dynamic network production structure. The proposed productivity indices consider multiple inputs (both discretionary and non-discretionary (quasi-fixed)), multiple desirable and undesirable inputs and outputs, and carryovers, all or some of which can take negative values. Further, the proposed models are feasible and bounded while computing cross-period DDFs. Our proposed *DSMLPI* is more advantageous because it is particularly suitable when firms are operating in an environment where technological innovations evolve rapidly, and there is no possibility of technical regress. The *DSMLPI* is based on the premise that production technology either progresses or at least stays unaltered. Two existing productivity indices, namely sequential Malmquist productivity index (*SMPI*) by Shestalova [86] and sequential Malmquist-Luenberger productivity index (*SMLPI*) by Oh and Heshmati [74], are also based on the same premise and preclude the possibility of technical regress. But, those indices treat the entire production process as a black-box and neglect the role of intermediate products or linking activities. Nevertheless, in developing *DSMLPI*, we not only consider the progressive feature of technology but also use a holistic representation of production technology that is based on a dynamic network structure, which recognizes the role of intermediate products in linking distinct divisions and carryover activities in linking two succeeding periods. This feature of representing production technology holistically using divisions or sub-processes in the productivity change measurement makes *DSMLPI* innovative and distinctive from *SMPI* and *SMLPI*. Therefore, we assert that *DSMLPI* provides more adequate productivity change estimates than any other standard black-box DEA-based productivity index. To the best of our knowledge, there is no programming-based TFP measurement model in the existing literature that can amalgamate all the above features. In this way, our paper offers a new endeavor in the TFP measurement altogether, especially in those situations when the production technology can be represented holistically using a dynamic network production structure, and there is no strict positivity in the data requirement. Hence, this paper significantly contributes to the existing literature on TFP measurement.

In addition to the modeling contribution, the empirical illustration is first to provide the risk-adjusted productivity estimates for Indian banks operating during the period from 2010 to 2017. The risk-adjusted productivity estimates are obtained by directly incorporating three internal risk control measures in the dynamic network production technology. Our three-stage production technology representation with deposit generation, lending, and revenue-generating processes not only resolves the most discussed “deposits dilemma”, but also delivers reliable productivity estimates by sufficiently incorporating risk control measures of equity capital, non-performing loans (NPLs), and loan loss provisions in the different stages of network production technology for a typical Indian bank. The equity capital, which accounts for both the bank’s risk-based capital requirements and the risk-return trade-off, is attuned as a quasi-fixed input in the deposit generation process. NPLs are unavoidable by-products of the lending process that occur alongside performing loans and raise capital expenses both in the short and long term. Fukuyama and Weber [42,43], Boussemart et al. [21], and Fukuyama and Tan [41] considered NPLs to be socially undesirable in the bank efficiency measurement. The dynamic network production structure used in this study recognizes

the undesirable character of NPLs and assigns a dual role to this risk control variable in the lending sub-process. In particular, we consider NPLs of the period $t - 1$ as an undesirable input in the lending process and NPLs of period t as the final undesirable output of the lending process. Since loan loss provisions directly affect the bank's profits, we include them as an additional choice input in the revenue-generating process.

The remainder of this paper unfolds as follows. Section 2 presents a very concise review of the literature on TFP measurement. Section 3 introduces the dynamic network production architecture and optimization models for computing *DMLPI* and *DSMLPI*. Section 4 presents an empirical illustration aiming to obtain the estimates of productivity change, efficiency change, and technical change for forty-two Indian banks operating during the period 2010–17. Section 5 finishes up the paper.

2. TFP growth measurement: a brief literature review

Attributable to the extensive use of TFP growth statistics as an economic indicator in policy modeling, researchers and practitioners remained motivated in productivity measurement since the seminal work of Solow [91] to develop and implement a large variety of models for TFP growth measurement. Earlier efforts to measure productivity change relied heavily on using the Divisia index of TFP growth, which can be computed directly from the observed data and is not dependent on any estimation. Nonetheless, an essential assumption of this index is that the underlying technology is homogeneous of degree one, making TFP growth coincides with technical change [56]. The eminent surveys by Fried et al. [39], Del Gatto et al. [26], and Sickles and Zelenyuk [87] archive the theoretical developments and empirical applications in the exciting field of TFP growth measurement. Among the existing approaches in the contemporary literature on the subject matter, both parametric and non-parametric frontier approaches to measure TFP growth have gained immense popularity due to their capability to use panel data and decomposing the TFP growth into distinct components such as efficiency change, scale effects, and technical change.

In the parametric frontier approach, one can make use of a battery of panel stochastic frontier models that are distinctive from one another and built on different functional forms of cost/profit/production/distance functions and distributional assumptions of error term representing (in)efficiency. To model the technical change component, a time trend variable is generally included in the right-hand side of the stochastic frontier models as a regressor. In this approach, empirical estimates of TFP growth and its components are obtained using the estimated parameter values and data. An excellent theoretical exposition of this approach is given in the famous book by Kumbhakar and Lovell [59], and the implementation of this approach using the Stata software is given in Kumbhakar et al. [58]. Over the years, numerous developments have emerged in SFA modeling framework. Sickles and Zelenyuk [87] document the complete journey of the SFA modeling from the first-generation models to the current second-generation models. The most recent advancements in SFA-based TFP growth measurement that have attracted the attention of practitioners is the work of Kumbhakar et al. [57] and Tsionas and Kumbhakar [95]. They introduced a SFA model, which is a four-way error component model. This model decomposes inefficiency into persistent (time-invariant/long-term) and transient (time-varying/short-term) components and separates these inefficiencies from unobserved firm effects (firm-heterogeneity) and random noise. Some notable applications of this SFA model to measure TFP growth estimates include Mattsson et al. [69] for the manufacturing industry, Badunenko and Kumbhakar [9] for banking firms, and Berisso [17] for agricultural farms.

The non-parametric frontier approach of TFP growth utilizes the DEA-based estimation of distance functions to compute various variants of the *MPI*. The most significant variants of *MPI* are the Adjacent-period *MPI* [36], Fixed-period *MPI* [14], Sequential *MPI* [86], Window *MPI* [7], Global *MPI* [79]. These variants differ on how the panel data is handled to specify the reference technology while computing contemporaneous and cross-period distance functions that are needed for obtaining yearly indices of TFP growth and its component measures. Simar and Wilson [88] developed a bootstrapping procedure for obtaining a confidence interval of *MPI*. Maniadakis and Thanassoulis [66] developed the cost *MPI* to identify allocative efficiency change. Tohidi et al. [93] proposed a global cost *MPI* which is circular and immune to infeasibility problem. The key advantage of *MPI* is no reliance on the functional form of the cost/profit/production/distance function and the distributional assumption of the error term capturing (in)efficiency. Some of the recent applications of *MPI* include [12,32,38,51,97], among others.

There are numerous theoretical extensions in conventional *MPI*. One of the most important extensions of *MPI* that pulled a great deal of interest from researchers is the Malmquist-Luenberger Productivity index (*MLPI*) proposed by Chung et al. [25]. Based on directional distance functions, *MLPI* measures environmentally sensitive productivity growth and its components in the presence of undesirable output(s). *MLPI* has found numerous applications (see, for example, [5,10,34,53,64]). In the existing literature, there are two popular extensions of *MLPI*. These extensions are the Global *MLPI* that was proposed by Oh [73] and the sequential *MLPI* that was developed by Oh and Heshmati [74]. The former extension overcomes the problem of infeasibility, while the latter extension considers the progressive nature of technology and rules out any possibility of spurious technical regress. It is noteworthy here that the standard *MPI*, *MLPI*, and their extensions consider the production technology as a “black-box” and do not account for any dynamic network production structure in which intermediate outputs link various sub-processes and carryover variables link time periods t and $t+1$.

Färe and Grosskopf [33] presented a discussion on dynamic structure with multiple inputs and outputs and developed network models. The survey article of Kao [50] documents theoretical developments in the network DEA modeling and reviews the recent empirical applications. In recent years, the dynamic network DEA approach received the attention of researchers because of its intrinsic advantage of opening the “black-box” of production into distinct sub-processes and using carryover variables to link the time periods in the process of efficiency evaluation. Tone and Tsutsui [94] formulated dynamic DEA models for network structure using the slack-based measure and applied their models to evaluate the performance of US electric companies over the year. In their survey articles, Fernández et al. [38] reviewed the applications of the dynamic network DEA approach in diverse fields.

3. Methodological framework

3.1. A general dynamic network structure and notations

An exposition of a dynamic network structure with K divisions and T periods of analysis for a DMU is given in Fig. 1. The multiple K divisions of a DMU are connected vertically via the intermediate products and horizontally across the consecutive periods through the carryovers between periods. In this paper, we assume that each division produces its outputs by consuming exogenous inputs and its carryovers from the previous period. An intermediate link between two divisions represents the outputs from the preceding division and the input for the succeeding division.

We used the following notations for building the *DMLPI* and *DSMLPI*.

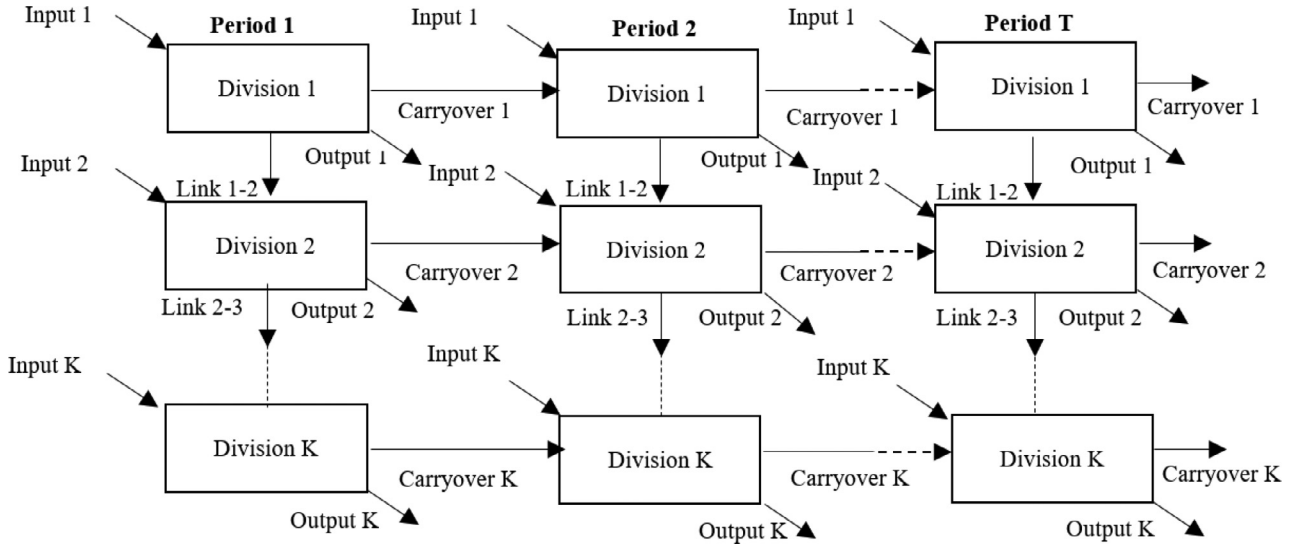


Fig. 1. A dynamic network series structure with K divisions and T periods of analysis.

Γ : time horizon of study.

$j \in \{1, \dots, n\}$: the index set of n DMUs.

$k \in \{1, \dots, K\}$: the index set of K divisions.

$t \in \{1, \dots, T\}$: the index set of T periods.

$i \in \{1, \dots, m\}$: the index set of m inputs.

$f \in \{1, \dots, q\}$: the index set of q quasi-fixed inputs.

$\ell \in \{1, \dots, \zeta\}$: the index set of ζ undesirable inputs.

$r \in \{1, \dots, s\}$: the index set of s desirable outputs.

$p \in \{1, \dots, \wp\}$: the index set of $\wp$ undesirable outputs.

$\ell_1 \in \{1, \dots, ngood\}$: the index set of $ngood$ desirable carryovers.

$g \in \{1, \dots, h\}$: the index set of h intermediate products.

o : a DMU under evaluation.

$X_{jk}^t = (x_{1jk}^t, \dots, x_{mjk}^t)'$: the vector of desirable inputs consumed by division k of DMU $_j$ in period t .

$V_{jk}^t = (v_{1jk}^t, \dots, v_{qjk}^t)'$: the vector of quasi-fixed inputs consumed by division k of DMU $_j$ in period t .

$W_{jk}^t = (w_{1jk}^t, \dots, w_{\zeta jk}^t)'$: the vector of undesirable inputs consumed by division k of DMU $_j$ in period t .

$Y_{jk}^t = (y_{1jk}^t, \dots, y_{s jk}^t)'$: the vector of desirable outputs produced by division k of DMU $_j$ in period t .

$U_{jk}^t = (u_{1jk}^t, \dots, u_{\wp jk}^t)'$: the vector of undesirable outputs produced by division k of DMU $_j$ in period t .

$C_{jk}^t = (c_{1jk}^t, \dots, c_{ngood jk}^t)'$: the vector of desirable carryovers produced by division k of DMU $_j$ from period t to $t+1$.

$Z_{jk}^t = (z_{1jk}^t, \dots, z_{h jk}^t)'$: the vector of intermediate products from division k to $k+1$ for DMU $_j$ in period t .

λ_{jk}^t : the intensity of division k for DMU $_j$ in period t .

We assume that the quasi-fixed input data is positive real while all other variables can take any real value.

3.2. Dynamic Malmquist-Luenberger productivity index

In this sub-section, we introduce a dynamic Malmquist-Luenberger productivity index (*DMLPI*), based on the contemporaneous efficiency frontier. To measure the productivity change net of the scale change between two consecutive periods t and $t+1$, the dynamic network production possibility set (PPS) in period t consuming the observations of period $t+1$ under the VRS assumption is defined as follows:

$$\begin{aligned}
 P^t = \{ & (X_k^{t+1}, V_k^{t+1}, W_k^{t+1}, Y_k^{t+1}, U_k^{t+1}, C_k^{t+1}) | X_k^{t+1} \geq \sum_{j=1}^n \lambda_{jk}^t X_{jk}^t, V_k^{t+1} \geq \sum_{j=1}^n (\lambda_{jk}^t + \phi_{jk}^t) V_{jk}^t, \\
 & W_k^{t+1} \leq \sum_{j=1}^n \lambda_{jk}^t W_{jk}^t, Y_k^{t+1} \leq \sum_{j=1}^n \lambda_{jk}^t Y_{jk}^t, U_k^{t+1} \geq \sum_{j=1}^n \lambda_{jk}^t U_{jk}^t, C_k^{t+1} \leq \sum_{j=1}^n \lambda_{jk}^t C_{jk}^t, \\
 & C_k^t \geq \sum_{j=1}^n \lambda_{jk}^t C_{jk}^{t-1}, k = 1, \dots, K, \sum_{j=1}^n \lambda_{jk}^t Z_{jk}^t = \sum_{j=1}^n \lambda_{jk+1}^t Z_{jk}^t, k = 1, \dots, K-1, \\
 & \sum_{j=1}^n \lambda_{jk}^t = 1, \lambda_{jk}^t, \phi_{jk}^t \geq 0, j = 1, \dots, n, k = 1, \dots, K \}
 \end{aligned} \tag{1}$$

where $X_k^t = (x_{1k}^t, \dots, x_{nk}^t)'$, $V_k^t = (v_{1k}^t, \dots, v_{nk}^t)'$, $W_k^t = (w_{1k}^t, \dots, w_{nk}^t)'$ are the vectors of desirable, quasi-fixed, and undesirable inputs for division k at time t , respectively; $Y_k^t = (y_{1k}^t, \dots, y_{nk}^t)'$, $U_k^t = (u_{1k}^t, \dots, u_{nk}^t)'$, $C_k^t = (c_{1k}^t, \dots, c_{nk}^t)'$, $Z_k^t = (z_{1k}^t, \dots, z_{nk}^t)'$ are desirable and undesirable outputs, desirable carryover, and intermediate link vectors for division k at time t , respectively.

We briefly explain the constraints for desirable carryover, intermediates, and quasi-fixed inputs in the formation of the PPS in (1). The constraints for the desirable carryover are built by taking motivation from the work of Tone and Tsutsui [94]. The desirable carryovers

are acting as desirable outputs from t and desirable inputs to $t+1$, which inspire to include two constraints corresponding to them. Again following Tone and Tsutsui [94], the linking activities are outputs from division k , and hence a larger amount is considered favorable, so

$$\sum_{j=1}^n \lambda_{jk}^t z_{gjk}^t \geq z_{gok}^t, g = 1, \dots, h, t = 1, \dots, T \quad (2)$$

Moreover, these activities are inputs to the division $k+1$, and hence a smaller amount is regarded favorable. Thus,

$$\sum_{j=1}^n \lambda_{jk+1}^t z_{gjk}^t \leq z_{gok}^t, g = 1, \dots, h, t = 1, \dots, T \quad (3)$$

From (2) and (3), we have

$$\sum_{j=1}^n \lambda_{jk}^t z_{gjk}^t \geq \sum_{j=1}^n \lambda_{jk+1}^t z_{gjk}^t, g = 1, \dots, h, t = 1, \dots, T \quad (4)$$

The constraint in (4) is imposed as the equality constraint in the PPS to ensure that there is no loss and no new creation of intermediate products while passing from division k to division $k+1$, $k = 1, \dots, K-1$.

Mehdiloo and Podinovski [71] developed the notion of hybrid production technology that allows us to model a group of closely related input and output measures in which some variables are both individually strongly disposable and weakly disposable. The constraints for the quasi-fixed inputs in division k at time t expressed in terms of the $n \times 1$ vectors λ_k^t and ϕ_k^t are inspired by this thought in the PPS P^t .

For $t = 1, \dots, T$, $k = 1, \dots, K$, define vectors $\mu_k^t = (\mu_{1k}^t, \dots, \mu_{mk}^t)'$, $\rho_k^t = (\rho_{1k}^t, \dots, \rho_{\zeta k}^t)'$, $v_k^t = (v_{1k}^t, \dots, v_{sk}^t)'$, $\omega_k^t = (\omega_{1k}^t, \dots, \omega_{\vartheta k}^t)'$, $\gamma_k^t = (\gamma_{1k}^t, \dots, \gamma_{ngoodk}^t)'$, and $\alpha_k^t = (\alpha_{1k}^t, \dots, \alpha_{ngoodk}^t)'$, with components:

$$\begin{aligned} \mu_{ik}^t &= \xi \times \max_{1 \leq j \leq n} \{ |x_{ijk}^t| \}, i = 1, \dots, m, \\ \rho_{\ell k}^t &= \min_{1 \leq j \leq n} \{ w_{\ell jk}^t \} - \pi, \ell = 1, \dots, \zeta, \\ v_{rk}^t &= \min_{1 \leq j \leq n} \{ y_{rjk}^t \} - \pi', r = 1, \dots, s, \\ \omega_{pk}^t &= \xi' \times \max_{1 \leq j \leq n} \{ |u_{pjk}^t| \}, p = 1, \dots, \vartheta, \\ \gamma_{\ell_1 k}^t &= \min_{1 \leq j \leq n} \{ c_{\ell_1 jk}^t \} - \pi'', \ell_1 = 1, \dots, ngood, \\ \alpha_{\ell_1 k}^t &= \xi'' \times \max_{1 \leq j \leq n} \{ |c_{\ell_1 jk}^t| \}, \ell_1 = 1, \dots, ngood, \end{aligned}$$

where $\xi, \xi', \xi'', \pi, \pi', \pi''$ are positive real constants, depending on t only, chosen appropriately as described later.

We propose a VRS model for evaluating the DDF $D^{t+a}(X^{t+b}, V^{t+b}, W^{t+b}, Y^{t+b}, U^{t+b}, C^{t+b}), a, b \in \{0, 1\}$, for DMU_o in period $t+a$ using the data of period $t+b$. This is described as follows:

$$(M1)_{a,b} \\ D^{t+a}(X^{t+b}, V^{t+b}, W^{t+b}, Y^{t+b}, U^{t+b}, C^{t+b}) = \max \beta_o^{a,b} \quad (5)$$

$$\text{s.t.} \sum_{j=1}^n \lambda_{jk}^{t+a} x_{ijk}^{t+a} \leq x_{iok}^{t+b} - \beta_o^{a,b} (x_{iok}^{t+b} + \mu_{ik}^{t+a}), i = 1, \dots, m, k = 1, \dots, K,$$

$$\sum_{j=1}^n (\lambda_{jk}^{t+a} + \phi_{jk}^{t+a}) v_{fjk}^{t+a} \leq v_{fok}^{t+b}, f = 1, \dots, q, k = 1, \dots, K, \quad (6)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} w_{\ell jk}^{t+a} \geq w_{\ell ok}^{t+b} + \beta_o^{a,b} (w_{\ell ok}^{t+b} - \rho_{\ell k}^{t+a}), \ell = 1, \dots, \zeta, k = 1, \dots, K, \quad (7)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} y_{rjk}^{t+a} \geq y_{rok}^{t+b} + \beta_o^{a,b} (y_{rok}^{t+b} - v_{rk}^{t+a}), r = 1, \dots, s, k = 1, \dots, K, \quad (8)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} u_{pjk}^{t+a} \leq u_{pok}^{t+b} - \beta_o^{a,b} (u_{pok}^{t+b} + \omega_{pk}^{t+a}), p = 1, \dots, \vartheta, k = 1, \dots, K, \quad (9)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} c_{\ell_1 jk}^{t+a} \geq c_{\ell_1 ok}^{t+b} + \beta_o^{a,b} (c_{\ell_1 ok}^{t+b} - \gamma_{\ell_1 k}^{t+a}), \ell_1 = 1, \dots, ngood, k = 1, \dots, K, \quad (10)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} c_{\ell_1 jk}^{t+a-1} \leq c_{\ell_1 ok}^{t+b-1} - \beta_o^{a,b} (c_{\ell_1 ok}^{t+b-1} + \alpha_{\ell_1 k}^{t+a-1}), \ell_1 = 1, \dots, ngood, k = 1, \dots, K, \quad (11)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} z_{gjk}^{t+a} = \sum_{j=1}^n \lambda_{j,k+1}^{t+a} z_{gjk}^{t+a}, g = 1, \dots, h, k = 1, \dots, K-1, \quad (12)$$

$$\sum_{j=1}^n \lambda_{jk}^{t+a} = 1, k = 1, \dots, K, \quad (13)$$

$$\beta_o^{a,b} \in \Re, \lambda_{jk}^{t+a}, \phi_{jk}^{t+a} \geq 0, j = 1, \dots, n, k = 1, \dots, K$$

where $X^{t+1} = (X_1^{t+1}, \dots, X_K^{t+1})'$, $V^{t+1} = (V_1^{t+1}, \dots, V_K^{t+1})'$, $W^{t+1} = (W_1^{t+1}, \dots, W_K^{t+1})'$, $Y^{t+1} = (Y_1^{t+1}, \dots, Y_K^{t+1})'$, $U^{t+1} = (U_1^{t+1}, \dots, U_K^{t+1})'$, $C^{t+1} = (C_1^{t+1}, \dots, C_K^{t+1})'$ are the desirable, quasi-fixed, and undesirable input, desirable and undesirable output, and desirable carryover vectors at time $t+1$, respectively. It is worth mentioning here that the data values are represented by small letters, while the entity or variable representing the entity is denoted by a capital letter. The emphasis of Model $(M1)_{a,b}$ is to evaluate the DDF depending on all types of inputs, outputs, and carryovers vectors, and hence we used the vector notations alongside D^{t+a} , subject to the constraints (5)-(13) to be satisfied by all data values that these vectors take for all $i, f, r, p, \ell_1, g, k, t$.

Note that while evaluating the cross-period DDFs $D^{t+a}(X^{t+b}, V^{t+b}, W^{t+b}, Y^{t+b}, U^{t+b}, C^{t+b})$, $a \neq b$, $a, b \in \{0, 1\}$, the constants $\xi, \xi', \xi'', \pi, \pi', \pi''$ are chosen such that $(x_{iok}^{t+b} + \mu_{ik}^{t+a})$, $(w_{\ell ok}^{t+b} - \rho_{\ell k}^{t+a})$, $(y_{rok}^{t+b} - v_{rk}^{t+a})$, $(u_{pok}^{t+b} + \omega_{pk}^{t+a})$, $(c_{\ell_1 ok}^{t+b} - \gamma_{\ell_1 k}^{t+a})$, $(c_{\ell_1 ok}^{t+b-1} + \alpha_{\ell_1 k}^{t+a-1})$ are positive, for all $i, \ell, r, p, \ell_1, k, t$.

Theorem 1. For $a, b \in \{0, 1\}$, model $(M1)_{a,b}$ is feasible and bounded, and its optimal objective value $\beta_{o^*}^{a,b} > -1$.

The proof is provided in Appendix A.

From this result, it follows that $1 + D^{t+a}(X^{t+b}, V^{t+b}, W^{t+b}, Y^{t+b}, U^{t+b}, C^{t+b}) > 0$, for $a, b \in \{0, 1\}$.

We are now ready to define the DMLPI of a DMU in the period t to $t+1$ as follows:

$$DMLPI^{(t,t+1)} = \left(\frac{1 + D^t(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + D^t(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \times \frac{1 + D^{t+1}(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + D^{t+1}(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \right)^{\frac{1}{2}}$$

which can be decomposed into efficiency change and technical change as follows:

$$\begin{aligned} DMLPI^{(t,t+1)} &= \left(\frac{1 + D^t(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + D^{t+1}(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \right) \\ &\times \left(\frac{1 + D^{t+1}(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})}{1 + D^t(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \times \frac{1 + D^{t+1}(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + D^t(X^t, V^t, W^t, Y^t, U^t, C^t)} \right)^{\frac{1}{2}} \\ &= EC^{(t,t+1)} \times TC^{(t,t+1)} \end{aligned} \quad (14)$$

where the efficiency change component $EC^{(t,t+1)}$ represents the movement of a DMU towards the best practice frontier (indicating a catch-up effect), and the technical change component $TC^{(t,t+1)}$ measures the shift in the efficient frontier composed by all divisions in period t to $t+1$ (indicating an innovation effect). In more simplified terms, the term $EC^{(t,t+1)}$ is related to the efforts put up by a DMU for improving its efficiency, while the term $TC^{(t,t+1)}$ reflects the change in the efficiency frontiers surrounding the DMUs between the two time periods.

Note that $DMLPI^{(t,t+1)} = 1$, $DMLPI^{(t,t+1)} > 1$, and $DMLPI^{(t,t+1)} < 1$ indicate no change, improvement, and regress, respectively, in the productivity of the DMU from period t to $t+1$.

3.3. Dynamic sequential Malmquist-Luenberger productivity index

In this sub-section, we introduce the dynamic sequential Malmquist-Luenberger productivity index (*DSMLPI*). The *DSMLPI* extends the Global Malmquist-Luenberger productivity index (*GMLPI*) of Emrouznejad and Yang [31] to the dynamic network production structure. The main strength of both *DSMLPI* and *GMLPI* is that these indices tackle the infeasibility problem effectively since the definitions of PPS coincide substantively with the sequential technology introduced by Tulkens and Vanden Eeckaut [96]. The sequential PPS at t establishes a reference production set formed by all the DMUs up to time t . It is defined by $\bar{P}^t = \text{conv}(P^1 \cup P^2 \cup \dots \cup P^t)$, $t = 1, \dots, T$, where P^t is the production technology of period t for dynamic network DEA and $\text{conv}(\cdot)$ denotes the convex hull. Hence, it assumes that the technologies of all the preceding periods are available in period t . The sequential production technology frontier for the dynamic DEA with network structure is described as follows:

$$\begin{aligned} \bar{P}^t &= \{(X_k^t, V_k^t, W_k^t, Y_k^t, U_k^t, C_{\ell_1 k}^t) | X_k^t \geq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} X_{jk}^{\tau}, V_k^t \geq \sum_{\tau=1}^t \sum_{j=1}^n (\lambda_{jk}^{\tau} + \phi_{jk}^{\tau}) V_{jk}^{\tau}, W_k^t \leq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} W_{jk}^{\tau}, \\ Y_k^t &\leq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} Y_{jk}^{\tau}, U_k^t \geq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} U_{jk}^{\tau}, C_k^t \leq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} C_{jk}^{\tau}, C_k^{t-1} \geq \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} C_{jk}^{\tau-1}, \\ \sum_{j=1}^n \lambda_{jk}^t Z_j^t &= \sum_{j=1}^n \lambda_{jk+1}^t Z_j^t, k = 1, \dots, K-1, \sum_{\tau=1}^t \sum_{j=1}^n \lambda_{jk}^{\tau} = 1, \\ \lambda_{jk}^t, \phi_{jk}^t &\geq 0, j = 1, \dots, n, k = 1, \dots, K, t = 1, \dots, T\}. \end{aligned} \quad (15)$$

Here $X_k^t = (X_{1k}^t, \dots, X_{nk}^t)'$, $V_k^t = (V_{1k}^t, \dots, V_{nk}^t)'$, $W_k^t = (W_{1k}^t, \dots, W_{nk}^t)'$, $Y_k^t = (Y_{1k}^t, \dots, Y_{nk}^t)'$, $U_k^t = (U_{1k}^t, \dots, U_{nk}^t)'$, $C_k^t = (C_{1k}^t, \dots, C_{nk}^t)'$, $Z_k^t = (Z_{1k}^t, \dots, Z_{nk}^t)'$ are the desirable, quasi-fixed, and undesirable inputs, desirable and undesirable outputs, desirable carryover, and intermediate link vectors for division k in period t , respectively.

Compared to the sequential PPS defined by Oh and Heshmati [74] for the conventional DEA models, our proposed definition includes the carryovers between two consecutive periods t and $t+1$ and also the intermediate products in the vertical links between different divisions.

For, $t = 1, \dots, T, k = 1, \dots, K$, define the vectors $\mu_k^t = (\mu_{1k}^t, \dots, \mu_{mk}^t)'$, $\rho_k^t = (\rho_{1k}^t, \dots, \rho_{\xi k}^t)'$, $v_k^t = (v_{1k}^t, \dots, v_{sk}^t)'$, $\omega_k^t = (\omega_{1k}^t, \dots, \omega_{\wp k}^t)'$, $\gamma_k^t = (\gamma_{1k}^t, \dots, \gamma_{ngoodk}^t)'$, $\alpha_k^t = (\alpha_{1k}^t, \dots, \alpha_{ngoodk}^t)'$ with respective components:

$$\begin{aligned}\mu_{ik}^t &= \eta \times \max_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ |x_{ijk}^\tau| \}, i = 1, \dots, m, \\ \rho_{\ell k}^t &= \min_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ w_{\ell jk}^\tau \} - \delta, \ell = 1, \dots, \zeta, \\ v_{rk}^t &= \min_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ y_{rjk}^\tau \} - \delta', r = 1, \dots, s, \\ \omega_{pk}^t &= \eta' \times \max_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ |u_{pjk}^\tau| \}, p = 1, \dots, \wp, \\ \gamma_{\ell_1 k}^t &= \min_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ c_{\ell_1 jk}^\tau \} - \delta'', \ell_1 = 1, \dots, ngood, \\ \alpha_{\ell_1 k}^t &= \eta'' \times \max_{1 \leq j \leq n, 1 \leq \tau \leq t} \{ |c_{\ell_1 jk}^\tau| \}, \ell_1 = 1, \dots, ngood.\end{aligned}$$

The positive constants $\eta, \eta', \eta'', \delta, \delta', \delta''$ depending on t only are chosen such that all direction vectors are positive and for all indices i, ℓ, r, p, ℓ_1 ,

$$\mu_{ik}^{t+1} < \mu_{ik}^t, \rho_{\ell k}^{t+1} > \rho_{\ell k}^t, v_{rk}^{t+1} > v_{rk}^t, \omega_{pk}^{t+1} < \omega_{pk}^t, \gamma_{\ell_1 k}^{t+1} > \gamma_{\ell_1 k}^t, \alpha_{\ell_1 k}^{t+1} < \alpha_{\ell_1 k}^t \quad (16)$$

For, $a, b \in \{0, 1\}$, the dynamic sequential DDFs are obtained by solving the following four linear programming problems:

$(M2)_{a,b}$

$$\begin{aligned}\bar{D}^{t+a}(X^{t+b}, V^{t+b}, W^{t+b}, Y^{t+b}, U^{t+b}, C^{dt+b}) &= \max \beta_o^{a,b} \\ \text{s.t. } \sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau x_{ijk}^\tau &\leq x_{iok}^{t+b} - \beta_o^{a,b} (x_{iok}^{t+b} + \mu_{ik}^{t+a}), i = 1, \dots, m, k = 1, \dots, K,\end{aligned} \quad (17)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n (\lambda_{jk}^\tau + \phi_{jk}^\tau) v_{fjk}^\tau \leq v_{fok}^{t+b}, f = 1, \dots, q, k = 1, \dots, K, \quad (18)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau w_{\ell jk}^\tau \geq w_{\ell ok}^{t+b} + \beta_o^{a,b} (w_{\ell ok}^{t+b} - \rho_{\ell k}^{t+a}), \ell = 1, \dots, \zeta, k = 1, \dots, K, \quad (19)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau y_{rjk}^\tau \geq y_{rok}^{t+b} + \beta_o^{a,b} (y_{rok}^{t+b} - v_{rk}^{t+a}), r = 1, \dots, s, k = 1, \dots, K, \quad (20)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau u_{pjk}^\tau \leq u_{pok}^{t+b} - \beta_o^{a,b} (u_{pok}^{t+b} + \omega_{pk}^{t+a}), p = 1, \dots, \wp, k = 1, \dots, K, \quad (21)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau c_{\ell_1 jk}^\tau \geq c_{\ell_1 ok}^{t+b} + \beta_o^{a,b} (c_{\ell_1 ok}^{t+b} - \gamma_{\ell_1 k}^{t+a}), \ell_1 = 1, \dots, ngood, k = 1, \dots, K, \quad (22)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau c_{\ell_1 jk}^\tau \leq c_{\ell_1 ok}^{t+b-1} - \beta_o^{a,b} (c_{\ell_1 ok}^{t+b-1} + \alpha_{\ell_1 k}^{t+a-1}), \ell_1 = 1, \dots, ngood, k = 1, \dots, K, \quad (23)$$

$$\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau z_{gjk}^\tau = \sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{j,k+1}^\tau z_{gjk}^\tau, g = 1, \dots, h, k = 1, \dots, K-1, \quad (24)$$

$$\begin{aligned}\sum_{\tau=1}^{t+a} \sum_{j=1}^n \lambda_{jk}^\tau &= 1, k = 1, \dots, K, \\ \beta_o^{a,b} \in \mathbb{R}, \lambda_{jk}^\tau, \phi_{jk}^\tau &\geq 0, j = 1, \dots, n, k = 1, \dots, K, \tau = 1, \dots, t+a\end{aligned} \quad (25)$$

where $X^{t+1} = (X_1^{t+1}, \dots, X_K^{t+1})'$, $V^{t+1} = (V_1^{t+1}, \dots, V_K^{t+1})'$, $W^{t+1} = (W_1^{t+1}, \dots, W_K^{t+1})'$, $Y^{t+1} = (Y_1^{t+1}, \dots, Y_K^{t+1})'$, $U^{t+1} = (U_1^{t+1}, \dots, U_K^{t+1})'$, $C^{t+1} = (C_1^{t+1}, \dots, C_K^{t+1})'$ are the desirable, quasi-fixed, and undesirable input, desirable and undesirable output, and desirable carryover vectors at time $t+1$, respectively.

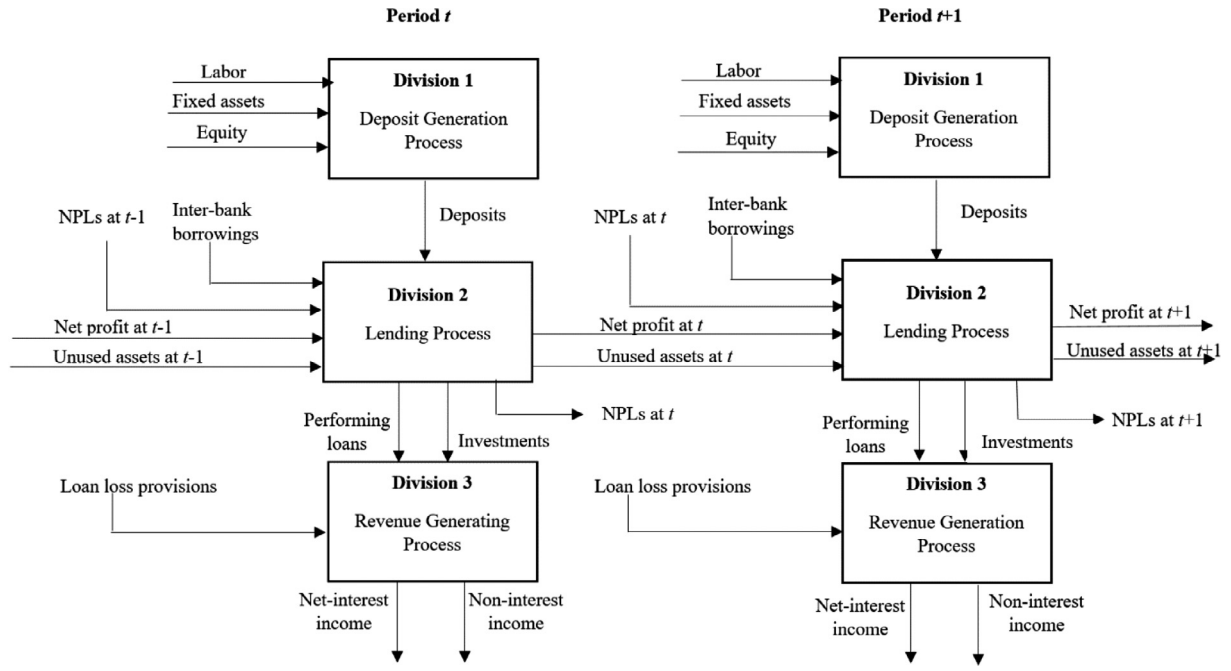


Fig. 2. The three-stage dynamic network structure for a typical bank.

Theorem 2. For $a, b \in \{0, 1\}$, model $(M2)_{a,b}$ is feasible and bounded, and its optimal objective value $\beta_{o^*}^{a,b} > -1$.

Proof can be worked out on the similar lines as proof of Theorem 1.

The DSMLPI between periods t and $t+1$ is defined by

$$DSMLPI^{(t,t+1)} = \left(\frac{1 + \bar{D}^t(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + \bar{D}^t(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \times \frac{1 + \bar{D}^{t+1}(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + \bar{D}^{t+1}(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \right)^{\frac{1}{2}}$$

The decomposition of DSMLPI into efficiency change and technical change is given as follows:

$$\begin{aligned} DSMLPI^{(t,t+1)} &= \left(\frac{1 + \bar{D}^t(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + \bar{D}^{t+1}(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \right) \\ &\times \left(\frac{1 + \bar{D}^{t+1}(X^t, V^t, W^t, Y^t, U^t, C^t)}{1 + \bar{D}^t(X^{t+1}, V^{t+1}, W^{t+1}, Y^{t+1}, U^{t+1}, C^{t+1})} \right)^{\frac{1}{2}} \\ &= \bar{EC}^{(t,t+1)} \times \bar{TC}^{(t,t+1)} \end{aligned} \quad (26)$$

where $\bar{EC}$ and $\bar{TC}$ capture the efficiency change and technical change between periods t and $t+1$, respectively. $\bar{EC} > (<)1$ implies that DMU_o shifts closer to (far away from) the efficient frontier period $t+1$ compared to period t . $\bar{TC}$ measures the shifting of an efficient frontier within the periods under consideration. The choice of the six constants (η s and δ s) in (16) ensures that $\bar{D}^{t+1}(X^\theta, V^\theta, W^\theta, Y^\theta, U^\theta, C^\theta) > \bar{D}^t(X^\theta, V^\theta, W^\theta, Y^\theta, U^\theta, C^\theta)$, $\theta = t, t+1$, it is not possible to obtain $\bar{TC} < 1$. Hence, $\bar{TC} > 1$ indicates technical progress while $\bar{TC} = 1$ means the technology remains unchanged from t to $t+1$. $DSMLPI^{(t,t+1)} = 1$ indicates no productivity change over two periods; $DSMLPI^{(t,t+1)} > 1$ depicts productivity growth while $DSMLPI^{(t,t+1)} < 1$ implies productivity regress from t to $t+1$.

4. Illustrated application

This section presents the empirical estimates pertaining to productivity change, efficiency change, and technical change indices for forty-two Indian banks that are obtained by implementing the DMLPI and DSMLPI frameworks. These indices are computed for the dataset ranged between 2010 and 2017 using the codes developed in LINGO.

Table 1

Inputs, outputs, intermediates, and carryovers for the period t used in the illustrated application

Desirable inputs of division 1	Labor
	Fixed assets
Quasi-fixed input of division 1	Equity
Link between divisions 1 and 2	Deposits
Desirable input of division 2	Inter-bank borrowings
Undesirable input of division 2	Non-performing loans (NPLs) at period $t-1$
Desirable carryovers of division 2	Unused assets at period $t-1$
	Net profit at period $t-1$
Links between divisions 2 and 3	Performing loans
	Investments
Undesirable output of division 2	Non-performing loans (NPLs) at period t
Desirable input of division 3	Loan loss provisions
Outputs of division 3	Net-interest income
	Non-interest income

Note: The variables net profit and net-interest income may take negative values.

Source: Authors' elaborations

4.1. Three-stage dynamic network production structure and variables

The three-stage dynamic network structure used in the illustrated application presents the underlined production process of a typical commercial bank operating in the Indian banking industry. Fig. 2 provides the detailed design of the underlined dynamic network structure for a bank. As can be seen from this figure, the production process of a bank within each period has three distinct divisions: i) deposit generation process, ii) lending process, and iii) revenue-generating process. These divisions are connected vertically with intermediate outputs and horizontally by carryover variables that link two distinct periods. Table 1 details the input, output, intermediate, and carryover variables used in our empirical illustration.

In division 1, which we call the deposit generation process, it is hypothesized that the bank uses two choice inputs (labor and fixed assets), and a quasi-fixed input (equity) to produce deposits, which is the sole output of division 1 and serves as an intermediate input in division 2. Two points are noteworthy here. First, to control for managerial risk preferences and the risk-return trade-off, we include equity as a quasi-fixed input that cannot be adjusted in the short-run. Some prominent bank efficiency and productivity studies that treat equity as a quasi-fixed input include Maudos et al. [70], Färe et al. [35], Girardone et al. [44], Yildirim and Philippatos [100], Bos et al. [20], Koutsomanoli-Filippaki et al. [52], and Gulati and Kumar [46,47]. Second, our treatment of deposits as an intermediate product addresses the popular “deposit dilemma” that is present in the bank efficiency literature due to disagreement between the intermediation and production approaches developed by Sealey and Lindley [85] and Benston and Smith [13], respectively, of selecting bank inputs and outputs in a study of bank performance. Here, we neither treat deposits as an input as proposed in the intermediation approach nor as an output as advocated in the production approach, but treat deposits as an intermediate output purely in lines recommended by Holod and Lewis [49].

In division 2, we hypothesize that the bank uses deposits as an intermediate product that originates in division 1 along with inter-bank borrowings (an additional choice input to this division) and non-performing loans (NPLs) of the previous period (as an undesirable input) to produce three outputs: i) performing loans (total loans minus non-performing loans), ii) investments, and iii) non-performing loans (as a final undesirable output). To connect the network structure of two adjacent periods, we use net profits and unused assets of the previous period as desirable carryovers, which affect the lending decision of the current period. Following Fukuyama and Weber [42], unused assets are measured as: to-

tal assets minus the sum of required reserves, fixed capital, loans, and securities investments. This division, which we call the lending process, captures the “intermediation efficiency” of the bank in the production process. The regulators are generally very keen to enhance the intermediation efficiency of the banks because most of their policy goals (like achieving high economic growth, provision of sufficient lending to small and medium enterprises, agriculture, and infrastructure projects, and so on, and the adequate funding needed for operationalizing state policies) can be achieved if banks are efficient in their intermediation activities.

There are two significant points about the lending process that merit due attention here. First, the lending decisions of the bank are constrained by the NPLs of the previous period. Hence, we treat NPLs of the period $t-1$ as an undesirable input in the lending process. Alongside traditional banking outputs (i.e., performing loans and investments), NPLs also originate as an undesirable output in the lending process due to irrational lending decisions. Such a dual role of NPLs in a network production structure can be found in the works of Akther et al. [4], Fukuyama and Weber [42,43], and Zha et al. [102]. Second, the inclusion of NPLs as an undesirable output in the lending process is entirely in line with contemporary research on banking performance. It is lauded that explicitly accounting for NPLs is indispensable for getting more meaningful efficiency and productivity measures (see, for example, [8,11,15,40,77,80,101]).

In division 3, which we call the revenue-generating process, the bank uses performing loans and investments as intermediate outputs alongside loan loss provisions (as an additional choice input to this division) to generate revenue in the form of net-interest income (interest earned minus interest expended) from its traditional lending activities and non-interest income from its non-traditional off-balance sheet activities. There is no doubt that the management of commercial banks always tries to maximize their efficiency in this division because this helps them to achieve their own target of profit maximization and financial sustainability. Three points are important to note here. First, we incorporate loan loss provisions as a choice input because these provisions are negatively associated with the banks' earnings. Earlier studies that include loan loss provisions as an input to control for risk while measuring the profit efficiency for banks include Drake [28], Drake and Hall [29], Drake et al. [30], and Pasiouras [78]. Second, NPLs emanating from the lending process (i.e., division 2) as an undesirable output are not treated as intermediate output for division 3 because of the reason that bad loans do not generate any interest income. Third, the non-interest income is equivalent to income from fee-based activities and correctly represents the non-traditional output of a bank. The inclusion of non-interest income as an output avoids the potential risk of misspecifying the banks' output and understating the level of bank efficiency and productivity. Gulati and Kumar [46] advocate the inclusion of non-interest income as a proxy for non-traditional activities in the model specification used for measuring the efficiency of Indian banks.

The contemporary literature on bank efficiency clearly highlights the importance of accounting for risk and asset quality (bad loans) in the assessment of efficiency levels and productivity changes for banks. In the above-presented three-stage dynamic network structure for Indian banks, we adequately incorporate risk and asset quality in the banks' production technology. Simper et al. [90] identified loan loss provisions, equity, and NPLs (as an undesirable output) as three risk management control variables. They inferred that the optimal input-output specification approach should include (any) two of the three risk control variables identified. The dynamic network model utilized here incorporates all the three risk control variables. Accordingly, it is expected that our dynamic network model will deliver consistent and meaningful risk-adjusted estimates of productivity growth and its compo-

Table 2
Summary statistics of variables used in the illustrated application.

Variables↓	Mean	σ	Kurtosis	Skewness	Median	Maximum	Minimum
Labor	11.43	4.62	4.72	2.11	9.96	28.10	6.70
Fixed assets	65.12	40.66	3.43	1.57	56.29	207.13	13.49
Equity	575.21	408.33	6.97	2.47	448.11	2207.89	150.66
Deposits	6734.06	2387.18	4.20	1.25	6277.40	15191.98	2466.53
Inter-bank borrowings	625.79	830.10	12.31	3.21	400.84	4389.92	2.94
NPLs	107.91	70.06	0.58	0.54	110.13	279.49	5.40
Unused assets	621.57	502.09	3.90	1.53	588.76	2395.39	64.75
Net profit	205.48	288.57	2.35	1.45	54.04	1064.63	-64.43
Performing loans	4872.88	1836.46	1.39	0.95	4584.40	10106.30	1683.45
Investments	2182.13	948.58	5.19	1.93	1984.79	5605.34	857.29
Loan loss provisions	97.94	72.49	4.69	1.45	87.57	313.39	-40.73
Net-interest income	205.40	100.10	3.23	1.76	186.46	525.27	68.83
Non-interest income	85.13	70.83	5.74	2.46	66.30	337.57	20.82

Note: All variables except labor are measured in terms of INR lakhs (10 lakhs=1 million).

Source: Authors' calculations.

nents. Overall, the three-stage dynamic network structure used in this empirical illustration unveils a very holistic presentation of the production technology used by commercial banks.

4.2. Data

For obtaining estimates of productivity change, efficiency change, and technical change for forty-two public and private banks over the study period 2010–2017, we culled out the required data on inputs, outputs, carryovers, and intermediates from the various issues of Statistical Tables Relating to Banks in India (an annual publication of the Reserve Bank of India). The study period encompasses the years after the recent global financial crisis of 2007–2009. The underlined sample period is very significant because, as far as we could possibly know, no study examined the temporal patterns of productivity growth of Indian banks in this period. The sampled banks are typically commercial banks that fulfill the conditions of Schedule II of the Reserve Bank of India Act, 1934. In public sector banks, the Government of India holds a majority stake (i.e., more than 50 percent), and these banks played a gargantuan role in the growth trajectory of the Indian economy. In the terminal year of the sampled period, the share of public sector banks in the Indian banking industry's deposits and advances are 72.7 percent and 68.5 percent, respectively. In recent years, private banks that are managed and controlled by private promoters strengthened their position in the Indian banking industry. This is all around reflected in their shares of 23.0 percent and 27.3 percent in the industry's deposits and advances, respectively, in the terminal sample year. As of now, private banks' market value is much higher than their public counterparts due to their innovative product offering, better management of their net interest margin, and relatively low NPLs. Indeed, even with the exclusion of foreign banks in our sample due to incomplete data on the majority of variables, we can safely infer that our sample is completely representative of the Indian banking industry considering the combined shares of sampled banks in key business parameters.

The current price figures of all variables (except labor) are deflated by the implicit price deflator of gross domestic product (GDP) at factor cost (with base 2011–12 = 100). Following Denizer et al. [27], we normalized all input, output, carryover, and intermediate variables of a bank by the number of branches owned by that bank. This normalization procedure requires dividing the values of all sorts of variables by the number of branches. The justification for using this normalization is that the normalization process limits the impact of potential measurement errors and reduces the data variations to a large extent. We assume that all banks under consideration are homogeneous in the sense that they all undertake similar activities and produce the same outputs using the

same inputs. Table 2 presents the descriptive statistics of all the variables used in the empirical illustration.

The descriptive statistics suggest notable differences in deposits, performing loans, unused assets, and investment across sampled banks. Further, a couple of sampled banks reported negative net profit values. Specifically, we note that 15 and 13 banks reported negative net profits in 2016 and 2017, respectively. However, the number of banks reporting a loss is relatively low for the remaining sample years. For instance, just 1 bank revealed a loss for 2010, 2012, and 2015; 2 banks observed a loss in 2014. Interestingly, no bank reported any loss in 2011 and 2013.

4.3. Results and discussion

4.3.1. Productivity change and its components

Appendix Tables B1 and B2 report the year-wise and bank-wise estimates of *DMLPI* and *DSMLPI* and their decomposed components: efficiency change and technical change. Based on these tables, we prepare Tables 3 and 4, which present geometric means of *DMLPI* and *DSMLPI* and their decomposed components for distinct ownership groups and individual sample banks. Following Lin and Chen [62], the Model $(M1)_{a,b}$ is solved by choosing the parameters $\xi = \xi' = \xi'' = 3$, $\pi = \pi' = \pi'' = 1$, for periods t and $t+1$. The Model $(M2)_{a,b}$ is solved by setting the parameters $\eta = \eta' = \eta'' = 3$, $\delta = \delta' = \delta'' = 1$, for period t , and $\eta = \eta' = \eta'' = 1.5$, $\delta = \delta' = \delta'' = 1$, for period $t+1$. We wish to underline here that the choice of these constants does not influence the overall inference of an empirical analysis; however, the values of *DMLPI*, *DSMLPI*, and their components may be affected slightly. Note that the geometric mean value of *DMLPI* (or *DSMLPI*) more than 1 indicates the productivity growth, equal to 1 infers productivity stagnation, and less than 1 signifies productivity regress. The same interpretation implies for decomposed components of *DMLPI* and *DSMLPI*. However, in the case of *DSMLPI*, it is not possible to obtain $\overline{TC} < 1$.

We first focus on the rows in bold in Table 3. Although empirical estimates of *DMLPI* and *DSMLPI* differ substantially (1.016 versus 1.031), both facilitate us to converge to an inference that the Indian banking industry experienced a total factor productivity growth during the sample period. Also, we find that the noticed productivity gain is essentially because of technical progress. The analysis of the correlation coefficients also upholds this finding. The correlation coefficient between $\overline{TC}$ and *DMLPI* is 0.999, and the same between $\overline{EC}$ and *DMLPI* is -0.801 for the banks under investigation. The correlation coefficient between $\overline{TC}$ and *DSMLPI* is 0.987, and the one between $\overline{EC}$ and *DSMLPI* is 0.214. The estimates of $\overline{TC}$ obtained through *DSMLPI* clearly show that technical regress is implausible when the banking units make use of all available

Table 3
Year-wise estimates of *DMLPI* and *DSMLPI* and their components

Productivity indices→	<i>DMLPI</i>			<i>DSMLPI</i>		
Year↓	<i>EC</i>	<i>TC</i>	<i>DMLPI</i>	$\bar{EC}$	$\bar{TC}$	<i>DSMLPI</i>
Panel A: All banks						
2011	1.003	0.988	0.991	0.987	1.012	0.999
2012	1.005	0.965	0.970	0.986	1.012	0.998
2013	0.999	1.152	1.151	0.996	1.143	1.138
2014	1.010	0.981	0.990	1.003	1.011	1.013
2015	1.002	1.006	1.009	1.005	1.006	1.011
2016	1.007	0.814	0.820	0.998	1.005	1.003
2017	0.998	1.235	1.232	1.006	1.058	1.064
2010-17	1.003	1.012	1.016	0.997	1.034	1.031
Panel B: Public sector banks						
2011	1.004	0.999	1.003	0.986	1.014	1.000
2012	1.008	0.982	0.989	0.983	1.015	0.998
2013	0.998	0.993	0.991	0.991	1.015	1.006
2014	1.013	0.991	1.003	1.004	1.006	1.010
2015	1.002	1.003	1.005	1.007	1.007	1.014
2016	1.010	0.970	0.980	0.999	1.007	1.006
2017	0.996	1.025	1.022	1.005	1.072	1.077
2010-17	1.004	0.995	0.999	0.996	1.019	1.015
Panel C: Private banks						
2011	1.003	0.970	0.973	0.988	1.010	0.998
2012	1.002	0.941	0.943	0.990	1.009	0.999
2013	1.000	1.435	1.434	1.002	1.361	1.364
2014	1.005	0.967	0.972	1.000	1.017	1.018
2015	1.003	1.012	1.014	1.001	1.005	1.006
2016	1.002	0.630	0.631	0.998	1.002	1.000
2017	1.001	1.622	1.624	1.008	1.037	1.045
2010-17	1.002	1.039	1.042	0.998	1.057	1.055
Panel D: Results of Li test						
H_0 : There is no difference in productivity change and its components in public and private banks						
Test statistics	37.858***	10.650***	12.935***	10.058***	-23.764*	-21.052**
p-value	0.000	0.000	0.000	0.000	0.071	0.041

Note: *, **, and *** indicate statistically significant at 10%, 5% and 1% levels, respectively.

Source: Authors' calculations

technology until the period under evaluation. This is evident from the fact that all computed yearly $\bar{TC}$ values are greater than one.

One of the key observations that come from the empirical estimates of productivity change and its decomposed components is that precluding the possibility of technical regress in *DSMLPI* yields higher estimates of productivity change and provides the most dependable estimates of efficiency change for banks. This observation is consistent for the industry as a whole and its segments defined on the basis of ownership. Note that the estimates of efficiency change indicating no catching-up effect in both public and private banking segments obtained from *DSMLPI* seem more reliable and convincing. The observed deterioration in the efficiency component of public sector banks, as indicated by *DSMLPI*, seems more consistent and goes precisely with the actual developments that happened in the banking landscape of India. In contrast, the evidence of efficiency improvement that *DMLPI* provides simply goes contrary. The various issues of Financial Stability Report and Report on Trend and Progress of Banking in India published by the Reserve Bank of India during the period 2010-17 well documented that public sector banks faced a tough time during the post-crisis period due to the rise in the gross non-performing assets (GNPAs) ratio from 2.26 percent in 2010 to 11.70 percent in 2017 [82]. This might be the prime reason for the efficiency decline in many banks during the study period. Consequently, to improve the operating efficiency of public sector banks, the Reserve Bank of India lended contaminated banks under the Prompt Corrective Action (PCA) Framework, and parallelly, the Indian government expressed its intention for a mega-merger and consolidation of these banks on many occasions since 2016. However, a formal announcement for the merger of 10 PSBs was made by Union Finance Minister Nirmala Sitharaman in August 2019.

Furthermore, higher estimates of technical progress and productivity growth yielded from *DSMLPI* seem plausible in light of the fact that Indian banks, irrespective of ownership structure, made heavy investments in technological innovations and electronic delivery channels. The most recent initiatives of the Indian government for a “cashless society” have additionally added a new impetus for banks to upgrade their technology levels. Overall, we find that the strong innovation effect due to positive developments in the technological front surged the total factor productivity levels even in the presence of no catching-up effect marked by observed efficiency decay because of the rise of default risk and low profitability levels.

The analysis of productivity change across ownership groups uncovers that private banks outperform their public counterparts (1.055 versus 1.015), and the primary contributor to their superior productivity performance is technological progress. The adoption of cutting-edge banking technology and persistent endeavors of private banks for technological advancement contributed towards their higher performance on the productivity front. We additionally find that, like public sector banks, the catching-up effect is also missing in private banks. This might be because of dwindling asset quality and rising expense-to-total income ratio in private banks during the post-crisis years. It is worth noting here that there was an ascent in the gross non-performing assets (GNPAs) ratio for private banks from 2.79 percent in 2010 to 4.20 percent in 2017. The results of the Li test [61], which compares the entire distributions of two samples, also confirm that the differences in productivity and its components across public and private banks are statistically significant (see Panel D of Table 3 for more details).

Table 4 reports the average *DMLPI* and *DSMLPI* and their components for individual banks over the period 2010-2017. We noted some striking differences in the estimates for productivity change,

Table 4Bank-wise estimates of *DMLPI* and *DSMLPI* and their components for Indian banks over the entire sample period (2010–2017)

Productivity indices→	<i>DMLPI</i>			<i>DSMLPI</i>		
Bank↓	<i>EC</i>	<i>TC</i>	<i>DMLPI</i>	<i>EC</i>	<i>TC</i>	<i>DSMLPI</i>
Panel A: Public sector banks						
State Bank of India	1.010	0.977	0.987	0.994	1.018	1.011
State Bank of Bikaner & Jaipur	1.006	0.993	0.999	0.995	1.007	1.002
State Bank of Hyderabad	1.006	0.991	0.997	0.993	1.013	1.006
State Bank of Mysore	1.007	0.994	1.001	0.996	1.013	1.009
State Bank of Patiala	1.002	0.992	0.994	0.995	1.011	1.006
State Bank of Travancore	1.007	0.993	1.000	0.999	1.012	1.011
Allahabad Bank	0.999	1.000	0.999	0.995	1.018	1.013
Andhra Bank	1.000	1.002	1.001	0.996	1.006	1.002
Bank of Baroda	1.000	1.010	1.010	1.010	1.036	1.046
Bank of India	1.011	0.986	0.998	0.998	1.059	1.057
Bank of Maharashtra	1.003	0.993	0.995	0.994	1.017	1.011
Canara Bank	1.002	0.993	0.996	0.999	1.016	1.015
Central Bank of India	1.003	0.999	1.002	1.000	1.010	1.010
Corporation Bank	1.003	0.999	1.001	0.994	1.021	1.015
Dena Bank	1.003	0.997	0.999	0.996	1.007	1.003
Indian Bank	0.999	0.997	0.996	0.997	1.005	1.002
Indian Overseas Bank	1.009	0.989	0.997	0.992	1.067	1.059
Oriental Bank of Commerce	1.004	0.995	0.999	0.995	1.032	1.027
Punjab & Sind Bank	0.998	1.001	0.999	0.997	1.004	1.002
Punjab National Bank	1.008	0.996	1.004	0.995	1.049	1.043
Syndicate Bank	1.007	0.992	1.000	0.993	1.008	1.001
UCO Bank	1.007	0.994	1.000	0.998	1.019	1.017
Union Bank of India	1.003	0.998	1.000	0.992	1.016	1.008
United Bank of India	1.004	0.991	0.996	0.998	1.015	1.013
Vijaya Bank	1.006	0.993	1.000	0.996	1.005	1.000
Geometric mean	1.004	0.995	0.999	0.996	1.019	1.015
Panel B: Private banks						
Catholic Syrian Bank	1.000	1.019	1.019	0.998	1.010	1.008
City Union Bank	1.002	1.024	1.026	0.999	1.030	1.029
Dhanalakshmi Bank	1.000	1.010	1.010	0.999	1.015	1.014
Federal Bank	1.008	1.023	1.031	0.996	1.022	1.018
Jammu & Kashmir Bank	1.001	1.073	1.074	0.998	1.070	1.069
Karnataka Bank	1.002	0.995	0.997	0.997	1.010	1.007
Karur Vysya Bank	1.005	1.024	1.029	0.996	1.023	1.018
Lakshmi Vilas Bank	1.003	1.006	1.009	0.997	1.025	1.022
Nainital Bank	1.000	1.070	1.070	1.006	1.069	1.076
Ratnakar Bank	1.000	1.070	1.070	1.001	1.039	1.040
South Indian Bank	1.003	1.047	1.050	1.004	1.043	1.047
Tamilnad Mercantile Bank	1.000	1.023	1.023	0.999	1.022	1.021
Axis Bank	1.000	1.072	1.072	1.006	1.101	1.107
HDFC Bank	1.000	1.079	1.079	0.995	1.106	1.100
ICICI Bank	1.002	1.008	1.010	0.990	1.195	1.183
Indusind Bank	1.001	1.108	1.108	1.009	1.120	1.129
Development Credit Bank	1.010	1.025	1.036	0.984	1.087	1.070
Geometric mean	1.002	1.039	1.042	0.998	1.057	1.055

Source: Authors' calculations

efficiency change, and technical change. First, the number of banks with efficiency decline is much higher when we apply the *DSMLPI* framework. In particular, we find that thirty-five out of forty-two banks observed an efficiency decline during the sample period when we utilize the *DSMLPI* approach. Out of seven sample banks experiencing a catching-up effect, five banks are private banks. However, only three banks with efficiency decline are noted using the *DMLPI* method. The difference in the number of efficient units by two indices can be explained as follows.

The *EC* and $\overline{EC}$ measure the shifting of a bank towards the efficient frontier composed by the production technology of period t utilizing the data of period $t + 1$. The convergence of a bank towards an efficient frontier occurs if either the PPS remains unchanged at $t + 1$ or expands but at a relatively lower rate than the rate at which the input-output bundle of a unit at $t + 1$ comes more proximate to the efficient frontier at t . However, if instead the PPS at $t + 1$ contracts then the unit can achieve the efficient frontier at period t without doing anything. This is one of the drawbacks of *DMLPI*. As $P^t \subseteq \overline{P}^t$, a large number of units can achieve the efficient frontier at t for *DMLPI* compared to *DSMLPI*,

and hence, $EC \geq \overline{EC}$. Second, we observe that spurious technical regress is likely to be observed when the *DMLPI* approach is applied; twenty-two banks (majority with public ownership) report a technological regress. The estimates of technical change obtained using the *DSMLPI* method rules out the possibility of any technical regress. This is evident from the fact that all banks in the sample show technical progress with $\overline{TC} > 1$, during the study period. Also, we note that there is no bank showing productivity regress in the *DSMLPI* approach, while seventeen banks are experiencing a productivity decline in the *DMLPI* approach. Overall, there are sharp differences in the estimates of productivity change and its components while using *DMLPI* and *DSMLPI* methods.

4.3.2. Comparison with standard Malmquist-Luenberger type productivity indices

We conceive that the proposed *DMLPI* and *DSMLPI* provide more adequate and dependable estimates of productivity change and its components than the standard versions of *MLPI* and *SMLPI* because these indices are based on the holistic representation of production technology rather than the “black-box” representation, which ig-

Table 5Year-wise estimates of *MLPI* and *SMLPI* and their components

Productivity indices→ Year↓	<i>MLPI</i>			<i>SMLPI</i>		
	<i>EC</i>	<i>TC</i>	<i>MLPI</i>	<i>EC</i>	<i>TC</i>	<i>SMLPI</i>
Panel A: All banks						
2011	1.019	0.989	1.008	0.992	1.105	1.096
2012	0.994	0.977	0.971	0.966	1.022	0.988
2013	0.966	1.022	0.988	0.948	1.045	0.991
2014	1.030	0.960	0.989	0.990	1.021	1.011
2015	1.008	0.976	0.984	0.981	1.036	1.017
2016	0.988	0.968	0.957	0.988	1.009	0.998
2017	0.986	1.026	1.011	1.004	1.025	1.029
2010-17	0.999	0.988	0.987	0.981	1.037	1.018
Panel B: Public sector banks						
2011	0.991	1.028	1.019	0.983	1.119	1.101
2012	0.991	0.960	0.952	0.949	1.017	0.965
2013	0.961	1.046	1.005	0.944	1.049	0.991
2014	1.018	0.969	0.986	0.977	1.023	1.000
2015	1.010	0.998	1.008	0.993	1.049	1.042
2016	0.972	0.973	0.946	0.971	1.013	0.983
2017	0.973	1.029	1.001	0.969	1.029	0.998
2010-17	0.988	1.000	0.988	0.970	1.042	1.011
Panel C: Private banks						
2011	1.060	0.935	0.991	1.004	1.085	1.089
2012	0.999	1.002	1.000	0.992	1.031	1.023
2013	0.975	0.987	0.963	0.955	1.038	0.991
2014	1.047	0.948	0.992	1.009	1.019	1.028
2015	1.005	0.946	0.950	0.963	1.018	0.980
2016	1.013	0.962	0.974	1.015	1.005	1.020
2017	1.006	1.021	1.027	1.058	1.018	1.077
2010-17	1.014	0.971	0.985	0.999	1.030	1.029

Source: Authors' calculations

nore the role of intermediates and carryover activities. To compute the standard (output-oriented) *MLPI* and *SMLPI* with observed input and output directions, we ignore all the stages of network production structure and utilize the inputs of division 1 and outputs of division 3 along with NPLs as additional undesirable outputs. Some prominent studies that explicitly include NPLs as an undesirable output in TFP estimation for banking firms include Park and Weber [77], Assaf et al. [8], Fujii et al. [40], and Zhu et al. [104]. Table 5 provides the relevant results.

We note that the estimated values of the standard *MLPI* are indicating the presence of productivity decline in the Indian banking industry during the sample period. Also, both public and private banks experience a productivity loss. Interestingly, private banks observe a technological regress while the technology remains stagnant in the public sector banks. The observed findings of *MLPI* are completely not convincing and do not reflect the true picture of the trajectory of productivity growth and the nature of technological progress in the Indian banking industry. We also note that the estimates for productivity change and its sources for the Indian banking industry provided by the *SMLPI* are lower than those obtained from *DSMLPI* (1.018 versus 1.031). This observation remains valid in the disaggregate analysis for ownership groups. On the basis of these findings, we can safely infer that ignoring the dynamic network structure in the productivity measurement framework understates the true productivity performance.

4.3.3. Second-stage regression analysis

To investigate the influencing factors of productivity growth and its sources in the Indian banking industry, we specify a dynamic panel model that takes the following form:

$$X_{jt} = \varphi_0 + \varphi_1 X_{j,t-1} + \sum_{k=2}^K \varphi_k \ln(Z^k)_{jt} + \varepsilon_{jt}, \quad \text{for } j = 1, \dots, n; t = 1, \dots, T.$$

where X_{jt} is the estimated value of *DMLPI* or *DSMLPI* and its component measures for j^{th} bank in the t^{th} year, Z^k_{jt} is a vector of k^{th} bank-specific variables that might affect the productivity growth and its distinct sources, and φ_s are the model parameters to be estimated. The recent literature on the two-stage approach of productivity analysis recognizes the problem of serial correlation as highlighted by Simar and Wilson [89] in efficiency and productivity measures estimated by DEA-based modeling frameworks like ours. Instead of using the computationally demanding bootstrapping approach to solve the serial correlation problem, which jeopardises the econometric estimation, Zhengfei and Oude Lansink [103] recommended the use of the system GMM model in the second-stage of productivity analysis. The compelling reasons for their recommendation in favor of the GMM approach are: i) this approach potentially handles the endogeneity and the serial correlation problems, and ii) this approach produces consistent parameter estimates and estimates of standard errors. Some notable studies that employed the GMM approach to identify key determinants of DEA-based productivity measures include Nakano and Managi [72]. Ojede et al. [76], Kurniawan and Managi [60], Wang and Wang, M. [98], Cao et al. [22], amongst others. In the present study, we apply the two-step system generalized method of moments (GMM) approach of Blundell and Bond [19] with Windmeijer's [99] correction.

Before presenting the econometric results of the dynamic panel modeling framework, we feel that a concise discussion on the selected exogenous bank-specific covariates included in the vector Z is warranted here and is given below. Growth in NPLs (G_NPLs) deteriorates the bank's asset quality and adversely affects bank efficiency and productivity growth [37,55]. Therefore, we hypothesize an inverse relationship between G_NPLs and bank productivity and its sources in line with the above argument. The natural logarithm of the bank's total assets is used to measure the variable *SIZE*. Mansour and Moussawi [67] construed a positive impact of bank size on productivity, implying the large-sized banks exploit economies of scale in the business operations, which may bring down the risk of defaults and improve their productivity level. Chortareas et al. [24] and Matthews et al. [68] held an opposite view and reported that the larger the bank size, the lower is the productivity growth. Hence, we hypothesize that *SIZE* may have either a positive or negative sign. The variable *IC* is measured by the ratio of intermediation cost to total assets and reflects the cost of financial intermediation of a bank. Berger and De Young [16] pointed out that the additional expenses incurred by the bank managers in controlling credit risk increase the operating costs by banks, bring down their efficiency levels, and backing the "bad management" hypothesis. In the Indian context, Kumar [54] found that extra cost incurred by the managers on the technology lowers the productivity levels. Consequently, we conjecture an inverse relation between *IC* and productivity indices. The variable *PRIORITY* is measured by a ratio of priority sector advances to total assets. Since a higher amount of directed lending at concessional rates under government-sponsored schemes adversely affects the bank earning potentials, we hypothesize a negative relationship between *PRIORITY* and bank productivity.

The variable *EQTA* is measured by the ratio of equity to total assets and is used as a proxy to gauge the influence of bank capitalization. The connection between bank capitalization and productivity is ambiguous in the previous literature. Mansour and Mousawi [67] observed that well-capitalized banks are more productive, while Mamatzakis and Tsionas [65] inferred that lower leverage in terms of higher *EQTA* might increase agency costs and reduce productivity growth. Hence, we have no *a priori* expectation on the sign of this variable. The variable *LIQUIDITY* speaks to the bank's liquidity position and is estimated by the ratio of loans to deposits. Greater liquidity improves the productivity growth of the

Table 6

Two-step system GMM estimates for the underlined dynamic panel data regressions.

Dependent variable→Regressors↓	EC	TC	DMLPI	EC	TC	DSMLPI
<i>One period lagged dependent variable</i>	-0.6724*** (0.2438)	-1.343*** (0.261)	-1.4585*** (0.2744)	0.5797** (0.263)	0.4094*** (0.139)	0.4220** (0.166)
<i>G_NPLs</i>	-0.0004 (0.0003)	-0.0231* (0.0317)	-0.0693* (0.04002)	0.0004 (0.0013)	-0.0035 (0.0071)	-0.0025 (0.007)
<i>SIZE</i>	0.0047 (0.0052)	-0.276 (0.311)	-0.7356 (0.5451)	0.0092 (0.014)	-0.1240 (0.1127)	-0.1104 (0.115)
<i>IC</i>	0.0021 (0.0037)	-0.0358 (0.2613)	-0.0415 (0.460)	-0.0260* (0.0141)	-0.0599 (0.1121)	-0.1124 (0.239)
<i>EQTA</i>	-	-5.4821 (5.1985)	-9.9701 (8.924)	0.5316** (0.0267)	-0.1450 (1.861)	0.6783 (2.348)
<i>LIQUIDITY</i>	-0.0240* (0.0133)	-	-	0.0286 (0.0278)	-	1.0179 (0.915)
<i>BME</i>	-0.0466 (0.0295)	-2.9302* (1.5397)	-5.2364 (3.3511)	-0.0216 (0.0532)	0.2058 (0.545)	1.4960* (0.833)
<i>PRIORITY</i>	-0.00003 (0.00026)	-0.0296** (0.0095)	-0.0273** (0.01282)	0.0007 (0.00071)	-0.0083* (0.0046)	-0.0018 (0.007)
<i>DIVERSIFICATION</i>	-0.0807*** (0.0288)	1.4109 (0.9811)	0.7385 (1.3457)	0.0631 (0.0505)	1.6274** (0.7441)	2.6636** (1.188)
Constant	1.7189*** (0.2388)	6.9432* (3.7537)	12.1912* (6.4616)	0.3123 (0.2785)	1.1299 (0.942)	-1.0843 (1.437)
AR(1) (p-value)	0.152	0.126	0.218	0.027	0.037	0.185
AR(2) (p-value)	0.135	0.055	0.743	0.729	0.214	0.188
Sargan (p-value)	0.018	0.490	0.979	0.950	0.000	0.789
Hansen J (p-value)	0.146	0.782	0.743	0.313	0.238	0.611
Wald Chi ²	491547***	462.02***	311.07***	54388***	6503.96***	4795.87***

Note: *, **, and *** indicate the significance of the relationship at 10%, 5% and 1% level, respectively.

Source: Authors' calculations.

banks. Therefore, we expect a positive link between *LIQUIDITY* and productivity growth. To analyze the impact of bank income diversification on its productivity performance, we follow Stiroh and Rumble [92] and define a diversification index (*DIVERSIFICATION*) based on the bank's share in net-interest and non-interest incomes. Sanyal and Shankar [83] reported an unfavorable impact of diversification on the productivity of banks. On the contrary, Ahamed [1] found that diversification benefits the banks in terms of higher productivity and profitability levels. Accordingly, the impact of income diversification on the productivity of banks isn't clear in the literature. Therefore, although we tend to anticipate a positive sign for this variable, it might also be negative. The variable *BME* represents bank monitoring effort, and the salaries-to-non-interest expenses ratio is used as a proxy for this variable. *BME* signifies the bank's efforts to improve staff quality, reflecting more labor expenses on effective loan monitoring which subsequently enhances productivity growth [3]. We expect that *BME* to have a positive impact on productivity growth and its components.

Table 6 reports the robust estimates of coefficients obtained using the two-step system GMM approach. Note that these estimates correspond to the best fitted regression equations based on statistical tests like the Arellano-Bond [6] autocorrelation test, Sargan test [84], and Hansen [48] test. We find that neither serial correlation nor invalid instruments pose any severe concern in the regression analysis. The results indicate that most of the estimated coefficients of *G_NPLs* have signs compatible with our *a priori* expectation. In particular, we find that growth in NPLs negatively and significantly influences *DMLPI* and *TC*. This finding facilitates us to derive an inference that the larger baggage of the bad loans of the bank significantly drags its innovation and productivity performance. The results also support the view that bank income diversification in terms of an increased share of non-interest income positively influences technological advancements and productivity change. However, in contrast, increased risk through greater income diversification by banks negatively impacts the efficiency change component. The finding that *IC* negatively affects *EC* is in line with our *a priori* expectation. The results also uncover a posi-

tive relation between *BME* and productivity growth, implying that the efforts in terms of improved monitoring and staff quality enhance the productivity levels of banks. Interestingly, the results of *EQTA* coefficients indicate that better capitalization enhances the only efficiency growth. Further, the impact of bank size is found to be statistically insignificant. Finally, we note that higher priority sector lending under government-sponsored schemes deteriorates the bank productivity growth.

5. Conclusions and future directions of research

The Malmquist-Luenberger productivity index (*MLPI*) has emerged as a standard tool to obtain the empirical estimates of productivity change in the presence of undesirable outputs. However, the infeasibility problem is widely pervasive in the standard version of *MLPI*. The present paper proposes two extensions of the conventional *MLPI*, which are capable of handling negative data values and successfully overcome the infeasibility problem in a production environment characterized by a general dynamic network production structure where multiple divisions are connected vertically via linked activities and multiple periods attached horizontally through carryover activities. We called these newly defined indices as the dynamic Malmquist-Luenberger productivity index (*DMLPI*) and dynamic sequential Malmquist-Luenberger productivity index (*DSMLPI*). These indices are based on the variable returns-to-scale (VRS) assumption and capture the productivity change net of the contribution of scale economies.

In spite of the fact that *DMLPI* provides us with an environmentally sensitive productivity growth incorporating undesirable outputs along with the desirable one, yet it fails to reflect the progressive nature of the production technology. The *DSMLPI* overcomes the key downside of *DMLPI* and precludes the possibility of any technical regress. Overall, the *DSMLPI* provides the most plausible productivity change estimates by eliminating the possibility of technical regress and contraction of production possibility set. We recommend the use of *DSMLPI* for analysis of productivity change with the perspective of medium or long-term periods where tech-

nology progresses rapidly. The empirical illustration based on Indian banking data features sharp contrasts in the estimates of productivity change, efficiency change, and technical change obtained using DMLPI and DSMLPI approaches. The differences in estimates stem primarily due to the imposition of different postulates while constructing these indices. The results presented in the empirical illustration unambiguously indicate that productivity change in the Indian banking industry is primarily driven by technical progress. This clearly highlights that Indian banks need to focus more on improving their technical efficiency. In light of the results obtained, we suggest that the management of banks should put more effort into improving the quality of the factors of production along with the allocation of these factors in the production process. The key finding of the second-stage regression analysis is that the growth in non-performing loans drags the productivity growth in the Indian banking industry. From a theoretical and practical perspective, future research can aim to develop a productivity index incorporating heterogeneity in production technology among the DMUs within a group. We can make the modeling more inclusive by including ordinal and interval data values.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Pooja Bansal: Writing – original draft, Writing – review & editing, Software, Methodology. **Sunil Kumar:** Conceptualization, Writing – original draft, Writing – review & editing, Formal analysis. **Aparna Mehra:** Writing – review & editing, Software, Methodology. **Rachita Gulati:** Writing – original draft, Data curation, Writing – review & editing, Formal analysis.

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Appendix A

Proof of Theorem 1 in Sub-section 3.2

Proof (i) If $a = b$, then model $(M1)_{a,b}$ is easily observed to be feasible by taking $\beta_0^{a,b} = 0$, $\lambda_{ok}^{t+a} = \lambda_{ok+1}^{t+a} = 1$, $\varphi_{jk}^{t+a} = 0$, $\forall j = 1, \dots, n$, and $\lambda_{jk}^{t+a} = \lambda_{jk+1}^{t+a} = 0$, $\forall j = 1, \dots, n$, $j \neq 0$. The optimal value $\beta_0^{a,a} \geq 0$.

(ii) Let $a \neq b$

The constraints (5) and (7)–(11) can be written as follows:

$$\left. \begin{aligned} \beta_0^{a,b} &\leq \frac{x_{iok}^{t+b} - \sum_{j=1}^n \lambda_{jk}^{t+a} x_{ijk}^{t+a}}{x_{iok}^{t+b} + \mu_{ik}^{t+a}}, i = 1, \dots, m, k = 1, \dots, K, \\ \beta_0^{a,b} &\leq \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} w_{ijk}^{t+a} - w_{iok}^{t+b}}{w_{iok}^{t+b} - \rho_{ik}^{t+a}}, \ell = 1, \dots, \zeta, k = 1, \dots, K, \\ \beta_0^{a,b} &\leq \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} y_{rjk}^{t+a} - y_{rok}^{t+b}}{y_{rok}^{t+b} - \nu_{rk}^{t+a}}, r = 1, \dots, s, k = 1, \dots, K, \\ \beta_0^{a,b} &\leq \frac{u_{pok}^{t+b} - \sum_{j=1}^n \lambda_{jk}^{t+a} u_{pjk}^{t+a}}{u_{pok}^{t+b} + \omega_{pk}^{t+a}}, p = 1, \dots, \wp, k = 1, \dots, K, \\ \beta_0^{a,b} &\leq \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} c_{1jk}^{t+a} - c_{1ok}^{t+b}}{c_{1ok}^{t+b} - \gamma_{\ell_1 k}^{t+a}}, \ell_1 = 1, \dots, ngood, k = 1, \dots, K, \\ \beta_0^{a,b} &\leq \frac{c_{1ok}^{t+b-1} - \sum_{j=1}^n \lambda_{jk}^{t+a} c_{1jk}^{t+a-1}}{c_{1ok}^{t+b-1} + \alpha_{\ell_1 k}^{t+a-1}}, \ell_1 = 1, \dots, ngood, k = 1, \dots, K \end{aligned} \right\} \quad (A)$$

We firstly intend to show that by choosing the six positive constants $\pi, \pi', \pi'', \xi, \xi', \xi''$ appropriately, the six ratios in the block (A) are greater than -1 .

From (7), for any $\pi > 0$,

$$\frac{\sum_{j=1}^n \lambda_{jk}^{t+a} w_{ijk}^{t+a} - w_{iok}^{t+b}}{w_{iok}^{t+b} - \rho_{ik}^{t+a}} \geq \frac{\min_{1 \leq j \leq n} \{w_{ijk}^{t+a}\} - w_{iok}^{t+b}}{w_{iok}^{t+b} - \rho_{ik}^{t+a}} > -1, \\ \ell = 1, \dots, \zeta, k = 1, \dots, K.$$

Similarly, for any $\pi' > 0$ and $\pi'' > 0$,

$$\frac{\sum_{j=1}^n \lambda_{jk}^{t+a} y_{rjk}^{t+a} - y_{rok}^{t+b}}{y_{rok}^{t+b} - \nu_{rk}^{t+a}} > -1, \quad \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} c_{1jk}^{t+a} - c_{1ok}^{t+b}}{c_{1ok}^{t+b} - \gamma_{\ell_1 k}^{t+a}} > -1.$$

Next, consider the ratio in the first inequality in the block (A).

We want, $x_{iok}^{t+b} + \mu_{ik}^{t+a} > 0$, and $\frac{x_{iok}^{t+b} - \sum_{j=1}^n \lambda_{jk}^{t+a} x_{ijk}^{t+a}}{x_{iok}^{t+b} + \mu_{ik}^{t+a}} > -1$,

hence, $\xi \max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\} > -x_{iok}^{t+b}$, and $\xi \max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\} > \max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\} -$

$$2x_{iok}^{t+b} \geq \sum_{j=1}^n \lambda_{jk}^{t+a} x_{ijk}^{t+a} - 2x_{iok}^{t+b}$$

We can thus choose

$$\xi > \max \left\{ 0, \max_{1 \leq i \leq m, 1 \leq k \leq K} \left\{ -\frac{x_{iok}^{t+b}}{\max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\}} \right\}, \right. \\ \left. \max_{1 \leq i \leq m, 1 \leq k \leq K} \left\{ \frac{\max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\} - 2x_{iok}^{t+b}}{\max_{1 \leq j \leq n} \{x_{ijk}^{t+a}\}} \right\} \right\}$$

By the similar arguments, we can choose ξ' and ξ'' , to ensure that the other two ratios in fourth and sixth inequalities in block (A) are greater than -1 .

Feasibility of the model:

The subsystem (12)–(13) is always consistent. Since the quasi-fixed data values are positive, we can appropriately choose $\varphi_{jk}^{t+a} \geq 0$, $\forall j, k, t$, to ensure consistency of (6). For any such feasible vector λ , we can choose $\beta_0^{a,b}$, as

$$-1 < \beta_0^{a,b} \min \left\{ \min_{1 \leq i \leq m, 1 \leq k \leq K} \left\{ \frac{x_{iok}^{t+b} - \sum_{j=1}^n \lambda_{jk}^{t+a} x_{ijk}^{t+a}}{x_{iok}^{t+b} + \mu_{ik}^{t+a}} \right\}, \min_{1 \leq \ell \leq \zeta, 1 \leq k \leq K} \left\{ \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} w_{ijk}^{t+a} - w_{iok}^{t+b}}{w_{iok}^{t+b} - \rho_{ik}^{t+a}} \right\}, \right. \\ \min_{1 \leq r \leq s, 1 \leq k \leq K} \left\{ \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} y_{rjk}^{t+a} - y_{rok}^{t+b}}{y_{rok}^{t+b} - \nu_{rk}^{t+a}} \right\}, \min_{1 \leq p \leq \wp, 1 \leq k \leq K} \left\{ \frac{u_{pok}^{t+b} - \sum_{j=1}^n \lambda_{jk}^{t+a} u_{pjk}^{t+a}}{u_{pok}^{t+b} + \omega_{pk}^{t+a}} \right\}, \\ \left. \min_{1 \leq \ell_1 \leq ngood, 1 \leq k \leq K} \left\{ \frac{\sum_{j=1}^n \lambda_{jk}^{t+a} c_{1jk}^{t+a} - c_{1ok}^{t+b}}{c_{1ok}^{t+b} - \gamma_{\ell_1 k}^{t+a}} \right\}, \min_{1 \leq \ell_1 \leq ngood, 1 \leq k \leq K} \left\{ \frac{c_{1ok}^{t+b-1} - \sum_{j=1}^n \lambda_{jk}^{t+a} c_{1jk}^{t+a-1}}{c_{1ok}^{t+b-1} + \alpha_{\ell_1 k}^{t+a-1}} \right\} \right\} \quad (27)$$

There exists a choice of $\beta_0^{a,b}$ as all six ratios in (27) are greater than -1 .

Optimal value of the model:

From (27), the feasible region of $(M1)_{a,b}$ is non-empty bounded above, hence possesses finite optimal value $\beta_0^{a,b} > -1$.

Appendix B

Table B1, Table B2.

Table B1

Year-wise estimates of *DMLPI*, *DSMLPI*, and its components for public sector banks

Year→	2011			2012			2013			2014			2015			2016			2017		
	DMLPI																				
Bank↓	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI
State Bank of India	1.003	1.005	1.007	1.022	0.942	0.963	0.983	0.982	0.966	1.035	0.965	0.998	1.012	0.999	1.011	1.025	0.931	0.955	0.988	1.019	1.007
State Bank of Bikaner & Jaipur	1.010	0.999	1.009	1.007	0.981	0.988	0.999	0.998	0.997	1.009	0.993	1.002	0.997	1.003	1.000	1.019	0.977	0.996	0.999	1.003	1.002
State Bank of Hyderabad	1.008	0.997	1.005	1.006	0.985	0.991	0.997	0.998	0.996	1.011	0.987	0.998	1.007	1.005	1.013	1.010	0.958	0.967	1.002	1.006	1.009
State Bank of Mysore	1.002	0.998	1.000	1.025	0.973	0.997	0.989	0.998	0.987	1.030	0.986	1.015	1.002	1.012	1.013	0.998	0.971	0.968	1.008	1.022	1.029
State Bank of Patiala	0.994	1.000	0.994	1.008	0.983	0.991	0.982	0.992	0.974	1.019	0.981	1.000	0.994	1.007	1.002	1.021	0.966	0.987	0.999	1.013	1.012
State Bank of Travancore	1.001	1.003	1.004	1.010	0.978	0.988	0.999	0.999	0.998	1.015	0.983	0.999	1.008	1.003	1.011	1.012	0.974	0.985	1.001	1.014	1.015
Allahabad Bank	1.005	0.996	1.001	1.008	0.994	1.002	0.990	0.997	0.987	1.011	0.999	1.010	0.997	1.011	1.008	1.004	0.973	0.977	0.982	1.031	1.012
Andhra Bank	0.999	1.000	0.999	1.000	1.002	1.002	0.999	1.001	1.000	1.005	0.993	0.998	0.994	1.006	1.000	1.012	0.986	0.998	0.989	1.024	1.013
Bank of Baroda	1.000	1.012	1.012	1.000	1.011	1.011	1.000	0.958	0.958	1.000	1.107	1.107	1.000	1.015	1.015	1.000	0.832	0.832	1.000	1.165	1.165
Bank of India	1.020	1.001	1.021	1.004	0.948	0.952	1.003	0.993	0.996	1.021	0.979	1.000	1.000	0.999	0.999	1.022	0.965	0.986	1.009	1.021	1.030
Bank of Maharashtra	1.003	0.999	1.002	1.008	0.990	0.998	1.000	0.988	0.988	1.005	0.967	0.971	1.006	0.996	1.002	1.010	0.985	0.995	0.988	1.025	1.013
Canara Bank	1.002	1.001	1.004	1.000	0.974	0.974	1.000	0.985	0.985	1.000	1.000	1.000	1.003	0.995	0.999	1.013	0.978	0.991	0.997	1.020	1.017
Central Bank of India	1.002	0.996	0.998	0.998	0.997	0.996	1.010	0.997	1.007	1.004	0.997	1.001	1.001	1.005	1.006	1.002	0.989	0.991	1.003	1.012	1.015
Corporation Bank	1.001	1.003	1.003	1.000	1.004	1.004	1.003	0.984	0.987	0.994	1.000	0.993	1.008	1.002	1.010	1.008	0.978	0.986	1.006	1.022	1.028
Dena Bank	1.004	0.999	1.003	1.000	0.997	0.997	1.000	0.995	0.995	1.009	0.985	0.994	1.008	0.999	1.007	1.001	0.982	0.984	0.995	1.020	1.015
Indian Bank	1.000	1.000	1.000	0.997	0.992	0.989	1.003	0.988	0.991	1.005	0.992	0.997	1.005	1.002	1.007	1.000	0.994	0.994	0.984	1.010	0.993
Indian Overseas Bank	1.005	0.992	0.997	1.032	0.912	0.941	0.980	0.999	0.979	1.035	0.981	1.015	0.995	1.007	1.002	1.015	0.987	1.002	1.000	1.047	1.047
Oriental Bank of Commerce	0.997	0.996	0.993	1.008	0.985	0.993	1.000	0.993	0.993	1.014	0.986	1.000	1.000	1.000	1.001	1.011	0.979	0.990	0.997	1.025	1.022
Punjab & Sind Bank	0.997	0.996	0.994	1.001	1.000	1.001	0.989	1.003	0.992	1.013	0.995	1.008	1.000	1.006	1.006	1.000	0.997	0.997	0.988	1.012	1.000
Punjab National Bank	1.007	0.999	1.006	1.010	0.983	0.993	1.014	1.001	1.014	0.996	0.982	0.978	1.004	1.002	1.006	1.021	0.978	0.999	1.009	1.027	1.036
Syndicate Bank	1.004	1.002	1.007	1.019	0.985	1.004	1.002	0.990	0.992	1.015	0.974	0.989	1.007	0.991	0.998	1.014	0.978	0.992	0.990	1.026	1.015
UCO Bank	1.016	0.998	1.014	1.007	0.963	0.970	0.995	0.998	0.993	1.017	0.998	1.014	0.999	0.999	0.997	1.006	0.979	0.985	1.008	1.022	1.030
Union Bank of India	1.006	0.998	1.005	0.995	0.995	0.991	1.003	1.001	1.004	1.024	0.976	1.000	1.001	1.003	1.004	1.016	0.980	0.996	0.973	1.030	1.002
United Bank of India	0.999	0.995	0.995	1.015	0.980	0.994	1.001	0.995	0.996	1.014	0.989	1.003	1.001	1.011	1.012	1.000	0.963	0.963	1.000	1.007	1.007
Vijaya Bank	1.003	0.999	1.001	1.008	0.988	0.996	1.000	0.994	0.994	1.017	0.983	1.000	1.006	0.992	0.997	1.014	0.977	0.990	0.996	1.022	1.018
	DSMLPI																				
	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI
State Bank of India	0.971	1.034	1.004	0.976	1.023	0.999	0.971	1.016	0.987	1.013	1.000	1.013	1.052	1.000	1.052	0.976	1.000	0.976	1.000	1.050	1.050
State Bank of Bikaner & Jaipur	0.987	1.017	1.004	0.994	1.011	1.005	0.983	1.018	1.001	0.999	1.003	1.003	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.003	1.003
State Bank of Hyderabad	0.994	1.010	1.004	0.972	1.022	0.994	0.982	1.025	1.007	1.005	1.000	1.005	1.000	1.010	1.010	1.000	1.000	1.000	1.000	1.023	1.023
State Bank of Mysore	0.974	1.020	0.994	0.968	1.025	0.992	0.999	1.015	1.014	1.014	1.002	1.016	1.014	1.004	1.018	0.993	1.004	0.996	1.009	1.024	1.033
State Bank of Patiala	0.981	1.008	0.989	0.977	1.016	0.993	0.972	1.026	0.998	0.998	1.003	1.001	1.000	1.016	1.016	1.016	1.005	1.021	1.022	1.000	1.022
State Bank of Travancore	0.977	1.024	1.000	0.988	1.016	1.003	0.974	1.028	1.002	1.001	1.004	1.005	1.009	1.008	1.017	0.995	1.003	0.999	1.050	1.001	1.051
Allahabad Bank	0.983	1.016	0.999	0.986	1.014	0.999	0.994	1.016	1.010	1.008	1.010	1.018	0.995	1.032	1.026	1.000	1.001	1.001	1.000	1.040	1.040
Andhra Bank	0.983	1.014	0.997	0.997	1.008	1.005	0.994	1.010	1.004	0.999	1.004	1.003	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.005	1.005
Bank of Baroda	1.000	1.007	1.007	1.000	1.009	1.009	1.000	1.006	1.006	1.000	1.064	1.064	1.000	1.000	1.000	1.000	0.975	1.047	1.021	1.022	1.125
Bank of India	1.006	1.012	1.018	0.956	1.029	0.984	1.012	1.010	1.021	1.013	1.009	1.021	0.994	1.013	1.007	1.024	1.048	1.072	0.985	1.329	1.308
Bank of Maharashtra	0.989	1.011	1.000	0.987	1.011	0.998	0.987	1.010	0.997	0.996	1.002	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.087	1.087
Canara Bank	1.000	1.007	1.006	1.001	1.002	1.004	0.984	1.006	0.990	1.006	1.006	1.012	0.998	1.002	1.000	1.000	1.000	1.000	1.002	1.094	1.096
Central Bank of India	0.981	1.014	0.994	0.987	1.013	1.000	1.009	1.006	1.015	1.000	1.000	1.000	1.016	1.000	1.016	0.994	1.004	0.998	1.014	1.030	1.044
Corporation Bank	0.994	1.014	1.008	0.997	1.006	1.004	0.968	1.017	0.985	0.996	1.010	1.005	1.000	1.005	1.005	1.000	1.000	1.000	1.005	1.097	1.103
Dena Bank	0.991	1.010	1.001	0.993	1.007	1.000	0.987	1.013	1.000	0.998	1.005	1.002	1.009	1.000	1.009	0.996	1.002	0.998	1.000	1.010	1.010
Indian Bank	1.000	1.003	1.003	0.978	1.005	0.984	0.990	1.020	1.010	1.006	1.003	1.009	1.007	1.000	1.007	0.999	1.001	1.000	0.999	1.000	1.000
Indian Overseas Bank	0.954	1.029	0.982	0.996	1.025	1.021	0.984	1.020	1.003	1.017	1.000	1.017	1.000	1.009	1.009	0.996	1.041	1.037	1.000	1.391	1.391
Oriental Bank of Commerce	0.985	1.004	0.989	0.980	1.018	0.998	1.010	1.022	1.033	1.000	1.008	1.008	0.989	1.025	1.014	0.999	1.000	1.000	1.000	1.154	1.154
Punjab & Sind Bank	0.989	1.004	0.992	0.991	1.006	0.997	0.987	1.013	1.000	1.008	1.004	1.012	1.013	1.000	1.013	0.994	1.004	0.998	1.000	1.000	1.000
Punjab National Bank	0.979	1.021	0.999	0.966	1.023	0.988	1.030	1.015	1.045	0.989	1.007	0.995	1.000	1.001	1.001	1.003	1.016	1.019	1.000	1.285	1.285
Syndicate Bank	0.986	1.021	1.007	0.981	1.017	0.997	0.987	1.013	0.999	1.001	1.002	1.003	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
UCO Bank	0.997	1.014	1.010	0.970	1.017	0.987	0.999	1.010	1.009	1.012	1.002	1.015	0.999	1.002	1.000	1.000					

Table B2
Year-wise estimates of DMLPI, DSMLPI, and its components for private banks

Year→	2011			2012			2013			2014			2015			2016			2017			
	DMLPI																					
Bank↓	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	EC	TC	DMLPI	
Catholic Syrian Bank	1.000	0.996	0.996	1.000	1.002	1.002	1.000	1.085	1.085	1.000	0.982	0.982	1.000	0.998	0.998	1.001	0.885	0.886	1.000	1.216	1.216	
City Union Bank	1.000	0.998	0.998	1.003	0.994	0.997	1.000	1.297	1.297	1.006	0.979	0.986	1.001	1.000	1.001	1.001	0.703	0.704	1.000	1.331	1.331	
Dhanalakshmi Bank	1.001	0.999	1.000	1.000	0.993	0.993	1.000	1.205	1.205	1.000	1.038	1.038	1.000	0.987	0.987	1.000	0.703	0.703	1.000	1.248	1.248	
Federal Bank	1.012	0.996	1.008	1.016	0.988	1.004	0.992	1.161	1.152	1.017	0.975	0.992	1.013	0.987	1.000	1.006	0.684	0.688	1.004	1.554	1.560	
Jammu & Kashmir Bank	1.000	1.003	1.003	1.006	0.998	1.003	1.000	1.546	1.546	1.000	0.967	0.967	1.000	0.998	0.998	1.000	0.659	0.659	1.000	1.668	1.668	
Karnataka Bank	0.997	0.998	0.995	1.016	0.989	1.004	1.000	1.024	1.024	1.000	0.998	0.998	1.000	1.000	1.000	1.000	0.736	0.736	1.001	1.304	1.306	
Karur Vysya Bank	1.000	1.007	1.007	0.998	0.983	0.981	1.000	1.179	1.179	1.031	0.970	1.000	1.004	0.995	0.998	1.002	0.790	0.792	1.001	1.322	1.324	
Lakshmi Vilas Bank	1.007	0.995	1.001	0.997	0.916	0.913	1.005	1.189	1.194	1.008	0.990	0.998	1.001	1.004	1.005	1.000	0.823	0.823	1.001	1.177	1.178	
Nainital Bank	1.000	0.996	0.996	1.000	1.000	1.000	1.000	1.637	1.637	1.000	0.986	0.986	1.000	0.998	0.998	1.000	0.494	0.494	1.000	2.031	2.031	
Ratnakar Bank	1.000	1.000	1.000	1.000	0.993	0.993	1.000	1.326	1.326	1.001	0.990	0.991	1.002	1.000	1.002	1.000	0.535	0.535	1.000	2.302	2.302	
South Indian Bank	1.003	0.998	1.000	1.003	0.995	0.997	1.000	1.431	1.431	1.011	0.971	0.981	1.001	0.999	1.001	1.001	0.654	0.654	1.002	1.535	1.539	
Tamilnad Mercantile Bank	1.000	1.001	1.001	1.000	0.991	0.991	1.000	1.204	1.204	1.001	0.987	0.988	1.001	1.000	1.000	1.000	0.850	0.851	1.000	1.169	1.169	
Axis Bank	1.000	0.875	0.875	1.000	0.925	0.925	1.000	1.817	1.817	1.000	0.918	0.918	1.000	1.078	1.078	1.000	0.439	0.439	1.000	2.550	2.550	
HDFC Bank	1.000	0.978	0.978	1.000	0.909	0.909	1.000	1.934	1.934	1.000	0.957	0.957	1.000	0.986	0.986	1.000	0.519	0.519	1.000	2.027	2.027	
ICICI Bank	1.000	0.722	0.722	1.000	0.639	0.639	1.000	2.180	2.180	1.000	0.870	0.870	1.000	1.075	1.075	1.014	0.442	0.449	1.000	2.535	2.535	
IndusInd Bank	1.002	1.005	1.007	1.000	0.984	0.984	1.000	2.084	2.084	1.000	0.928	0.928	1.000	1.111	1.111	1.000	0.418	0.418	1.002	2.301	2.307	
Development Credit Bank	1.029	0.980	1.009	1.000	0.794	0.794	1.000	1.769	1.769	1.013	0.941	0.953	1.022	0.996	1.017	1.004	0.666	0.669	1.003	1.385	1.389	
	DSMLPI																					
	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	EC	TC	DSMLPI	
Catholic Syrian Bank	0.991	1.002	0.993	1.005	1.002	1.007	1.000	1.037	1.037	1.000	1.003	1.003	1.003	1.003	1.004	1.008	1.001	1.002	1.003	0.988	1.015	1.003
City Union Bank	0.994	1.002	0.996	0.993	1.004	0.997	1.002	1.202	1.205	1.000	1.009	1.009	1.011	1.000	1.011	0.995	1.003	0.999	0.999	1.004	1.004	
Dhanalakshmi Bank	0.997	1.003	1.000	0.994	1.003	0.997	1.004	1.133	1.138	1.000	1.038	1.038	1.012	0.917	0.928	0.997	1.007	1.003	0.991	1.014	1.004	
Federal Bank	0.986	1.014	1.000	0.984	1.013	0.997	0.984	1.115	1.097	1.013	1.004	1.017	1.000	1.000	1.000	1.000	1.000	1.000	1.002	1.015	1.017	
Jammu & Kashmir Bank	0.996	1.008	1.004	0.997	1.008	1.005	1.001	1.427	1.429	0.994	1.008	1.002	1.000	1.000	1.000	1.000	1.003	1.003	1.000	1.098	1.098	
Karnataka Bank	0.980	1.009	0.988	0.990	1.011	1.001	1.005	1.032	1.038	1.000	1.005	1.005	1.010	1.000	1.010	0.994	1.003	0.997	1.001	1.009	1.010	
Karur Vysya Bank	1.000	1.008	1.008	0.966	1.010	0.975	0.994	1.138	1.131	1.013	1.005	1.018	0.998	1.001	0.999	1.000	1.001	1.001	0.999	1.005	1.003	
Lakshmi Vilas Bank	0.979	1.011	0.990	0.995	1.012	1.006	1.009	1.126	1.137	1.000	1.016	1.016	1.000	1.008	1.008	0.996	1.002	0.998	0.999	1.005	1.004	
Nainital Bank	1.000	1.001	1.001	1.000	1.000	1.000	1.000	1.509	1.509	1.000	1.035	1.035	1.010	1.004	1.014	1.003	1.003	1.005	1.029	1.014	1.043	
Ratnakar Bank	1.000	1.001	1.000	0.985	1.004	0.989	0.999	1.237	1.235	1.003	1.013	1.017	1.004	1.000	1.004	0.996	1.000	0.997	1.017	1.038	1.055	
South Indian Bank	0.998	1.003	1.001	0.994	1.003	0.997	1.004	1.314	1.319	1.000	1.006	1.006	1.002	1.001	1.003	0.997	1.002	0.998	1.035	1.006	1.041	
Tamilnad Mercantile Bank	1.000	1.003	1.003	0.985	1.004	0.989	1.005	1.138	1.144	1.000	1.008	1.008	1.004	1.001	1.005	0.997	1.002	0.999	1.002	1.002	1.004	
Axis Bank	1.000	1.005	1.005	1.000	1.001	1.001	1.000	1.764	1.764	0.997	1.025	1.022	1.053	1.004	1.058	0.990	1.000	0.990	1.000	1.073	1.073	
HDFC Bank	1.000	1.008	1.008	1.000	1.001	1.001	1.000	1.832	1.832	1.000	1.042	1.042	0.946	1.032	0.976	1.000	1.000	1.000	1.018	1.017	1.035	
ICICI Bank	0.944	1.040	0.983	1.002	1.019	1.021	1.010	2.237	2.260	1.000	1.041	1.041	0.972	1.049	1.020	1.000	1.003	1.003	1.000	1.343	1.343	
IndusInd Bank	1.002	1.010	1.012	0.998	1.001	0.999	1.001	1.963	1.965	1.000	1.027	1.027	0.999	1.073	1.072	1.000	1.000	1.000	1.062	1.009	1.072	
Development Credit Bank	0.932	1.046	0.975	0.949	1.057	1.003	1.021	1.603	1.637	0.988	1.009	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.005	1.005	

Source: Authors' calculations

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